

Shape, kinematics, and cores of cosmologically motivated dry major merger remnants with supermassive black hole binaries

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

Stellar cores in massive elliptical galaxies may record the action of supermassive black hole (SMBH) binaries, but reconstructing how much stellar mass these binaries eject remains challenging: the original density profile is unknown, and the galaxy merger itself redistributes stars. We investigate this ambiguity using high-resolution N -body simulations of 28 cosmologically motivated, low-redshift dry major mergers, with progenitor structures and orbits drawn from ILLUSTRISTNG. Matched simulations with and without SMBHs isolate the binaries’ contribution to central stellar depletion. The initial encounter eccentricity governs the remnants’ global structure: more radial mergers produce more prolate, more slowly rotating and more radially anisotropic galaxies, with flatter central profiles. SMBH binaries leave the global structure largely unchanged, but enlarge cores and make the centre less prolate and tangentially anisotropic. Because the progenitors already have shallow central profiles, core flattening alone is a poor diagnostic of binary scouring. Core radii ($R_b \simeq 0.2\text{--}2$ kpc) scale nearly linearly with the binary influence radius as $R_b \propto r_{\text{inf}}^{1.03 \pm 0.13}$, and increase with binary mass as $R_b \propto M_{\text{bin}}^{0.66 \pm 0.19}$. The inferred ejected mass ($0.25\text{--}1.7 M_{\text{bin}}$) increases approximately linearly with binary mass and rises with encounter eccentricity. Conventional Sérsic-based mass deficits can substantially overestimate this mass and correlate only weakly with it ($R^2 = 0.38$). Combining the mass deficit with stellar velocity dispersion, effective radius and inner density slope improves the fit to $R^2 = 0.75$, providing an observationally accessible estimator of stellar ejection and a firmer basis for interpreting the signatures of SMBH binaries in galaxy cores.

Key words: black hole physics – gravitational waves – methods: numerical – galaxies: kinematics and dynamics – galaxies: interactions – galaxies: nuclei.

1 INTRODUCTION

Early-type galaxies exhibit a pronounced dichotomy in their structural and kinematic properties. The most luminous ellipticals are generally slowly rotating systems with boxy isophotes and shallow central stellar profiles, whereas lower luminosity systems tend to show greater rotational support, disk isophotes, and steeper central profiles (Kormendy & Bender 1996; Faber et al. 1996; Cappellari 2016). These differences are thought to reflect distinct evolutionary pathways. While less massive early-type galaxies can retain signatures of dissipative processes and gas rich mergers (Barnes & Hernquist 1996; Cappellari et al. 2007; Hopkins et al. 2008; Johansson et al. 2009; Cappellari 2016), the subsequent growth of the most massive ellipticals below redshifts $z \lesssim 2\text{--}3$ is expected to be dominated by relatively dry, dissipationless mergers (Naab et al. 2009; Oser et al. 2010; Feldmann et al. 2011; Moster et al. 2013; Rodríguez-Gómez et al. 2016). In this regime, the structure of the remnant is determined primarily by the dynamics of the merger rather than by radiative cooling and star formation.

Galaxy mergers do not simply increase the mass of their progenitors: their orbital configuration and mass ratio can leave a substantial imprint on the properties of the remnant. Numerical experiments

have shown that the remnant shape (Villumsen 1982; Boylan-Kolchin et al. 2005; González-García & Van Albada 2005a,b; Boylan-Kolchin et al. 2006; Li et al. 2018; Bortolas et al. 2018), amount of rotational support (González-García & Van Albada 2005a,b; Moody et al. 2014; Bortolas et al. 2018; Li et al. 2018) and velocity anisotropy (González-García & Van Albada 2005a,b; Bournaud et al. 2007; Bortolas et al. 2018) can depend on the orbital parameters of the encounter. The structure and kinematics of the resulting galaxy can therefore vary substantially for different merger configurations.

A further defining feature of massive ellipticals is the presence of central stellar cores. Over most of the radial extent of elliptical galaxies, the surface brightness profile is well described by a Sérsic model (Sérsic 1963; Caon et al. 1993), yet the most massive systems frequently show a central flattening relative to the inward extrapolation of this profile (Lauer et al. 2007; Dullo & Graham 2014). These cores typically extend over tens to hundreds of parsecs and can reach kiloparsec scales in the most extreme systems (Ravindranath et al. 2002; Lauer et al. 2007; Rusli et al. 2013; Dullo & Graham 2014; Dullo et al. 2017; Dullo 2019).

A natural explanation for the formation of these cores is the *scouring* of stars by supermassive black hole (SMBH) binaries formed during galaxy mergers. Following the merger of their host galaxies, the two SMBHs sink towards the centre through dynamical friction and eventually form a bound binary. Subsequent interactions with

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surrounding stars extract energy and angular momentum from the binary through three-body encounters, ejecting stars from the central region and progressively flattening the stellar density profile (Begelman et al. 1980; Ebisuzaki et al. 1991; Makino & Ebisuzaki 1996; Quinlan & Hernquist 1997; Milosavljević & Merritt 2001; Merritt 2006; Rantala et al. 2018, 2019). This interpretation is supported, first, by evidence for a coevolution between SMBHs and their host galaxies, as indicated by correlations between black hole mass and global bulge properties, most notably stellar velocity dispersion and stellar mass (Magorrian et al. 1998; Ferrarese & Merritt 2000; Gebhardt et al. 2000; Tremaine et al. 2002; Häring & Rix 2004). There are also observational indications that the size of galaxy cores and the amount of stellar mass apparently missing from their centres scale with the mass of the central SMBH (Graham 2004; de Ruiter et al. 2005; Lauer et al. 2007; Kormendy & Bender 2009; Rusli et al. 2013; Dullo & Graham 2014; Thomas et al. 2016). Together, these correlations have naturally motivated the idea that the central structure of massive ellipticals may retain a dynamical imprint of SMBH evolution.

This is particularly relevant given that SMBH binaries formed during galaxy mergers are expected to emit gravitational waves in the nanohertz regime accessible to Pulsar Timing Arrays (Agazie et al. 2023a; EPTA Collaboration et al. 2024). Understanding the interaction between SMBH binaries and their stellar environments is therefore important both for interpreting the observed properties of galaxy nuclei and for modelling the gravitational wave signals produced during SMBH binary evolution.

However, these correlations do not by themselves establish that SMBH binaries are responsible for the observed cores. The pre-existing stellar density profile of a galaxy is generally unknown, making it difficult to determine whether an observed core was produced by stellar depletion by an SMBH binary or instead reflects a pre-existing shallow central profile. Moreover, the commonly used mass deficit is not a direct measurement of the mass removed from the nucleus: it depends on the assumed reference profile and can vary substantially between different fitting procedures (Hopkins & Hernquist 2010; Dullo & Graham 2013, 2014). The interpretation of observed core properties therefore depends on understanding of how the initial galaxy structure and the merger itself determine the subsequent evolution of the nucleus.

A number of theoretical studies have investigated this problem using a range of idealised and increasingly realistic models. For example, Merritt (2006) found, using a spherical equilibrium galaxy model, that the stellar mass deficit produced by a binary scales with the binary mass and, in repeated merger models, can increase cumulatively with the number of mergers. Subsequent simulations, however, found that neither the core size nor the mass deficit grows linearly with the number of successive mergers (Nasim et al. 2021; Dosopoulou et al. 2021). Other scenarios have also been proposed to produce large depleted cores, including recoiling SMBHs, the heating of the central stellar distribution by an infalling massive perturber, interactions involving multiple SMBHs, and the combined effects of SMBH evolution and AGN feedback (Boylan-Kolchin et al. 2004; Merritt et al. 2004; Gualandris & Merritt 2008; Goerdt et al. 2010; Kulkarni & Loeb 2012; Martizzi et al. 2012; Nasim et al. 2021; Khonji et al. 2024).

The development of high-resolution galaxy merger simulations has made it possible to follow SMBH binaries from the large-scale merger down to their interaction with the central stellar population. Rantala et al. (2018), for example, followed complete dry mergers with a high-resolution regularised N -body treatment and demonstrated that the SMBH mass and the progenitor inner density slope affect the

resulting core size, surface density profile, and velocity anisotropy. Dosopoulou et al. (2021) subsequently revisited core formation in a systematic study, exploring the dependence on the progenitor density slope, SMBH binary mass ratio, and morphology of the merger remnant. Their results confirmed the lack of a linear growth of core size and mass deficit with the number of successive mergers found by Nasim et al. (2021), while showing that the ejected stellar mass depends only weakly on the binary mass ratio and initial inner density slope, and that the evolution of the core properties is strongly influenced by the remnant morphology. While these studies substantially improved upon earlier idealised treatments, the progenitor galaxies in both cases were constructed from idealised analytical models rather than being drawn directly from cosmological galaxy populations. Their parameter explorations therefore provide limited information on how the correlations between progenitor properties, merger orbits, and remnant properties arise in a realistic cosmological context.

In this work, we use high-resolution collisionless N -body simulations with initial conditions extracted from the ILLUSTRISTNG cosmological simulations to study massive, low redshift, dry major merger remnants hosting SMBH binaries relevant to the PTA signal. This cosmological selection provides a broad range of progenitor properties and merger orbits, allowing us to investigate how these initial conditions shape the resulting galaxies and their central structures. We further derive empirical scaling relations to identify the progenitor and merger properties that best predict the properties of the remnants and the stellar mass ejected by the SMBH binary. The remainder of this paper is organised as follows:

- **Section 2:** We describe the methodology adopted in this work, including the selection of mergers from ILLUSTRISTNG, the extraction of their initial conditions, and the setup of the high-resolution N -body simulations. We then introduce the measurements used to characterise the remnant shape, kinematics, and core properties, followed by the procedure used to derive the scaling relations.
- **Section 3:** We present our results, first examining how the orbital properties of the cosmological galaxy mergers are reflected in the shape and kinematics of the remnants, and then turning to their central core properties. We derive scaling relations and investigate both the mutual correlations between the remnant properties and their dependence on the merger and progenitor properties.
- **Section 4:** We summarise our main findings, place them in the context of previous work, and discuss the main caveats of our study.

2 METHODS

2.1 Initial conditions from cosmological simulations

IllustrisTNG. We extract the initial conditions for our merger simulations from the ILLUSTRISTNG suite of large volume cosmological magnetohydrodynamical simulations, performed with the moving mesh code AREPO (Springel 2010). The suite comprises three simulation volumes, TNG50, TNG100, and TNG300, with box sizes of 51.7, 110.7, and 302.6 cMpc, respectively (Springel et al. 2018; Pillepich et al. 2018b; Naiman et al. 2018; Nelson et al. 2018; Marinacci et al. 2018; Nelson et al. 2019a,b; Pillepich et al. 2019). The simulations follow the coupled evolution of dark matter, gas, stars, and SMBHs from $z = 127$ to $z = 0$ within a Λ CDM cosmology consistent with the Planck 2015 results (Planck Collaboration et al. 2016). SMBHs are seeded with an initial mass of $8 \times 10^5 h^{-1} M_{\odot}$ in dark matter haloes above $5 \times 10^{10} h^{-1} M_{\odot}$ (Pillepich et al. 2018a) and grow via gas accretion following a Bondi–Hoyle prescription,

limited by the Eddington threshold. They remain anchored to the minimum of their host potential and are assumed to merge promptly once the corresponding merger criterion is satisfied (Weinberger et al. 2017, 2018). We use the highest-resolution realizations of each volume, TNG50-1, TNG100-1, and TNG300-1, which contain 2×2160^3 , 2×1820^3 , and 2×2500^3 resolution elements, respectively. Further details on the ILLUSTRISTNG simulations are provided in François et al. (2026); see also Weinberger et al. (2017); Pillepich et al. (2018a) for detailed descriptions of the simulation methodology.

Merger selection. We construct our merger sample by selecting systems representative of the massive, low redshift galaxy mergers expected to contribute significantly to the PTA signal. We begin by identifying, at $z = 0$, galaxies with stellar masses $M_\star > 3 \times 10^{11} M_\odot$ in TNG50, TNG100, and TNG300, resulting in 33, 149, and 1552 galaxies, respectively. Their assembly histories are then reconstructed using the SUBLINK merger trees (Rodríguez-Gomez et al. 2015). We identify major mergers with stellar mass ratios greater than 1/4 occurring at $z \leq 1$, motivated by the expectation that low redshift major mergers provide an important contribution to the stochastic GWB (Sesana et al. 2008; Izquierdo-Villalba et al. 2022; Agazie et al. 2023b). For each candidate merger, we determine the orbital parameters of the two progenitors and exclude encounters on unbound orbits. This initial selection yields 256 galaxy mergers. The properties of the progenitor galaxies are measured in the snapshot immediately preceding the merger.¹

We next require both progenitors to host a SMBH and impose the additional conditions $q_{\text{BH}} > 1/4$ and $M > 10^8 M_\odot$ on the resulting SMBH binary (Sesana et al. 2009; Izquierdo-Villalba et al. 2022; Agazie et al. 2023b), where

$$M = \frac{(M_1 M_2)^{3/5}}{(M_1 + M_2)^{1/5}}, \quad (1)$$

M_1 and M_2 denote the masses of the two black holes and q_{BH} is the BH mass ratio. These criteria reduce the sample to 74 mergers.

Finally, we restrict the sample to systems for which the gas content is sufficiently low for our collisionless N -body treatment to provide a reasonable description of the subsequent evolution. Specifically, we require the gas mass within twice the stellar half-mass radius to be less than 30% of the stellar mass. This final cut leaves 30 mergers. However, the subsequent SMBH binary evolution presented in François et al. (2026) showed that two of these highly eccentric systems form SMBH binaries at very high eccentricity and coalesce almost immediately thereafter, leaving little time for significant stellar scouring. We therefore exclude these two systems from the present analysis, resulting in a final sample of 28 mergers.

Extraction of initial conditions. We reconstruct the structural properties of the progenitor galaxies from their ILLUSTRISTNG density profiles. For both the stellar and dark matter components, we fit a truncated spheroidal profile of the form

$$\rho(r) = \rho_0 \left(\frac{r}{r_s} \right)^{-\gamma} \left[1 + \left(\frac{r}{r_s} \right)^\alpha \right]^{(\gamma-\beta)/\alpha} \exp \left[- \left(\frac{r}{r_{\text{cut}}} \right)^\xi \right], \quad (2)$$

using a Markov chain Monte Carlo (MCMC) procedure, where we fix $\alpha = 1$ and $\beta = 3$, while ρ_0 , r_s , γ , r_{cut} , and ξ are left free but

¹ If one progenitor is not identified in this snapshot, we trace the merger trees backwards until both progenitors are simultaneously identified; systems for which this is not possible, or that no longer satisfy $q_\star > 1/4$, are discarded. See François et al. (2026).

constrained to be positive. The exponential truncation allows the outer profile to account for the substantial stripping experienced by progenitors in the cosmological environment. Examples of the stellar and dark matter density profile fits are shown in Fig. 1.

We characterise the orbital configuration of each encounter by treating the two progenitors as point masses in a Keplerian two-body system. Using their relative separation r_{rel} and velocity v_{rel} , we compute the semi-major axis and eccentricity of the galaxy encounter as

$$a_0 = \left(\frac{2}{r_{\text{rel}}} - \frac{v_{\text{rel}}^2}{GM} \right)^{-1}, \quad e_0 = \sqrt{1 - \frac{l^2}{GMa_0}}, \quad (3)$$

where M is the total mass enclosed within the relative separation, G is the gravitational constant, and l is the specific angular momentum. The corresponding pericentre and apocentre are given by

$$r_{\text{peri}} = a_0(1 - e_0), \quad r_{\text{apo}} = a_0(1 + e_0). \quad (4)$$

The relative velocity is computed after subtracting the bulk motions of the progenitors, measured within their respective stellar half-mass radii. We refer to François et al. (2026) for a detailed description of the extraction procedure and the resulting distributions of the initial conditions.

2.2 High-resolution N -body simulations

2.2.1 Particle sampling and mass resolution

Sampling. The stellar and dark matter components of each progenitor are resampled from the fitted ILLUSTRISTNG density profiles using the AGAMA library (Vasiliev 2019). We construct self-consistent, action-based distribution functions from the fitted density profiles and their corresponding gravitational potentials, assuming spherical symmetry, and draw phase-space samples from these distributions. The resulting progenitors are initially isotropic, with no imposed rotational support. This procedure allows us to reconstruct the progenitors at substantially higher resolution than in ILLUSTRISTNG.

Number of particles. The total number of particles is chosen individually for each merger, balancing the need for high central mass resolution against the computational cost of the simulations. This is particularly important for our highly eccentric encounters, for which numerical fluctuations in the central stellar distribution can affect the binary eccentricity at formation, e_b (Gualandris et al. 2026). We therefore aim to maximise the initial number of particles of the first progenitor within five times the binary influence radius, while keeping the total integration time computationally tractable. This results in typically $(4-12) \times 10^5$ particles within this region. The first progenitor is sampled with equal numbers of stellar and dark matter particles, and the particle masses are chosen such that the two progenitors have the same particle masses while preserving their mass ratio. The total number of particles is typically in the range $(4-12) \times 10^6$.

Mass refinement. We further employ a mass refinement scheme to increase the mass resolution in the central regions. Particles are oversampled, ranked according to their pericentric distance and divided into four shells, with the innermost shell retaining the highest mass resolution and progressively larger particle masses are assigned to outer shells. The same procedure is applied independently to the stellar and dark matter components, resulting in eight particle populations. The adopted mass multipliers and population

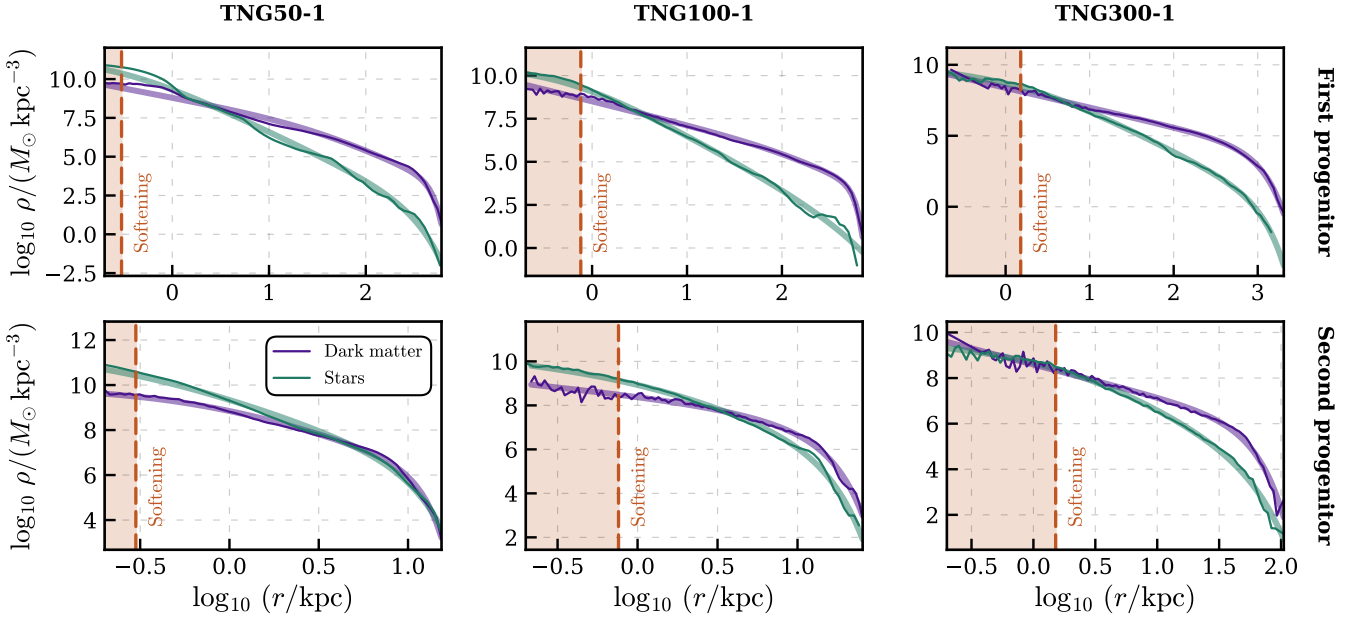


Figure 1. Density profile fits for galaxies extracted from the ILLUSTRISTNG simulations. The thin, dark curves show the original TNG density profiles, while the thick, lighter curves represent the corresponding best fitting truncated spheroidal models. The dark matter and stellar components are indicated in purple and green, respectively. Each column shows an example pair of galaxies selected from one of the three ILLUSTRISTNG volumes, with the first and second progenitors shown in the top and bottom rows, respectively. Adapted from François et al. (2026).

Table 1. Parameters specifying the mass-refinement scheme adopted in our simulations. For each shell, we report the relative particle mass multiplier and the fraction of the total particle population assigned to that shell.

Shell index	mass multiplier	fraction of population
1	1	0.05
2	3	0.45
3	10	0.35
4	40	0.15

fractions are given in Table 1, with the total mass and density profiles preserved by the refinement. The resulting central particle masses are of order a few $10^4 M_\odot$. This configuration provides high mass resolution in the region relevant for the SMBH binary while keeping the masses of outer shell particles sufficiently low. We refer to François et al. (2026) for the complete list of particle numbers adopted for each simulation, as well as for the resolution tests and more details on the mass refinement scheme.

Orbital insertion. The two progenitors are placed on the Keplerian orbit inferred from their ILLUSTRISTNG properties, using the corresponding semi-major axis a_0 and eccentricity e_0 . We initialise all simulations at the apocentre of the reconstructed orbit. This choice allows the galaxies to undergo tidal interactions during the initial approach to their first pericentric passage.

2.2.2 *N*-body code and softening

We follow the collisionless dynamical evolution of the merger with GRIFFIN, a high-performance *N*-body code based on the fast multipole method (FMM; Dehnen 2014). The FMM is used to compute the gravitational forces between stellar and dark matter particles,

with a computational scaling of $O(N^{0.87})$. Interactions involving the SMBHs, however, are evaluated by direct summation to ensure accurate treatment of their mutual dynamics and close encounters. The code continuously monitors the force errors and adaptively adjusts its numerical parameters, providing a controlled and well behaved error distribution comparable to those obtained with direct summation methods.

Because our simulations contain particle populations spanning a range of masses, we use mass dependent gravitational softenings for the stellar and dark matter components. More massive particles are assigned larger softenings, thereby reducing spurious collisional effects such as artificial mass segregation and the associated sinking of massive particles.

The softening length for a particle of mass m is given by

$$\varepsilon = \varepsilon_0 \left(\frac{m}{m_0} \right)^{1/2}, \quad (5)$$

where $\varepsilon_0 = 20$ pc is the reference softening length and m_0 is the mass of the lightest, most centrally refined dark matter particles. The resulting softenings for stellar particles are typically ~ 2 – 3 pc to 20 pc, while those of dark matter particles range from ε_0 to ~ 125 pc for the most massive particles. The $\varepsilon \propto m^{1/2}$ scaling ensures that close encounters between particles remain sufficiently softened to preserve the collisionless character of the system.²

The SMBHs are treated separately, with a constant softening during the early dynamical friction dominated evolution, followed by a smaller softening once a bound binary is about to form. We adopt $\varepsilon_{\text{BH,df}} = 10$ pc during the initial evolution and reduce it to $\varepsilon_{\text{BH,bin}} = 2$

² This scaling is motivated by requiring the acceleration generated by close particle–particle encounters to remain well below that of the smooth mean field potential.

pc after the third pericentric passage, shortly before the formation of the bound SMBH binary.

2.3 Computation of shape, kinematics and core properties

2.3.1 Shape

To characterise the global shape of the galaxies in ILLUSTRISTNG and of our merger remnants, we first rank the particles according to their binding energy and retain the 50% most bound particles. We then compute the shape tensor of this subset,

$$S_{ij} = \frac{\sum_k m_k (r_k)_i (r_k)_j}{\sum_k m_k}, \quad (6)$$

where m_k and $(r_k)_i$ are the mass and the i -th coordinate of particle k , respectively.³ Diagonalising S yields the principal axes of the ellipsoid that best describes the stellar distribution, with eigenvalues ordered such that $a > b > c$. The corresponding axis ratios, $p \equiv b/a$ and $q \equiv c/a$, quantify the departure from spherical symmetry, for which $p = q = 1$.

To improve the accuracy of the measurement, we iteratively recompute the shape using only particles enclosed within the ellipsoid defined by the previous iteration, until the axis ratios have a fractional change between iterations $\leq 10^{-3}$. The final values of p and q fully specify the intrinsic shape of the stellar distribution. We further quantify its triaxiality using

$$T = \frac{1 - p^2}{1 - q^2}. \quad (7)$$

The triaxiality parameter lies in the range $0 \leq T \leq 1$ and provides a convenient classification of the remnant shape. Values $T < 0.5$ correspond to predominantly oblate configurations, $T = 0.5$ to a maximally triaxial shape, and $T > 0.5$ to predominantly prolate configurations.

We additionally characterise the intrinsic shape of the central stellar distribution using the same procedure. Rather than selecting the 50% most bound particles, we consider two central regions defined by characteristic radii: the SMBH binary influence radius, r_{inf} , and $r_{5\%}$, the radius enclosing 5% of the total stellar mass of the remnant.

2.3.2 Kinematics

We characterise the rotational support of the stellar component using the ratio v_φ/σ_φ , where v_φ is the mean azimuthal velocity and σ_φ is the corresponding velocity dispersion. The z -axis is aligned with the total angular momentum of the selected stellar particles. We also compute the one-dimensional (1D) velocity dispersion, defined as $\sigma = \sqrt{(\sigma_r^2 + \sigma_\theta^2 + \sigma_\varphi^2)/3}$, where σ_r , σ_θ , and σ_φ are the velocity dispersions in the radial, polar, and azimuthal directions, respectively. The velocity anisotropy of the stellar component is quantified by the parameter β , defined as $\beta = 1 - (\sigma_\theta^2 + \sigma_\varphi^2)/2\sigma_r^2$. For $\beta = 0$, the velocity distribution is isotropic, while $\beta < 0$ and $\beta > 0$ correspond to tangentially and radially biased velocity distributions, respectively.

³ The shape tensor S is related to the moment of inertia tensor I through $I = \text{tr}(M_{\text{sub}}S)\mathbf{1} - M_{\text{sub}}S$, where $\mathbf{1}$ is the identity tensor and M_{sub} is the total mass of the selected particles.

2.3.3 Core properties

Projections. Because the merger remnants are not perfectly spherical, their projected core properties can depend on the viewing direction. We therefore measure the core properties for several projections separately and use the median across the projections as our fiducial value, thereby mitigating the dependence on the viewing direction. Each projection is defined by a point on the unit sphere, whose position specifies the direction normal to the corresponding projection plane. For a triaxial system with reflection symmetry about its principal axes, the eight octants of the unit sphere provide equivalent sets of viewing directions. We therefore restrict the sampling to a single octant and select 12 uniformly distributed directions using the Fibonacci method (González 2010). This construction provides a more homogeneous coverage of the sphere than independently sampling the polar and azimuthal angles uniformly, which would result in an artificial concentration of directions near the poles. The resulting set of projections provides a representative sampling of the range of viewing angles relevant for the projected core properties.

Core-Sérsic fit. We first estimate the core radius and inner density slope by fitting a core-Sérsic model to the projected stellar surface density profile of each of the 12 projections. Following Graham et al. (2003), we adopt the functional form

$$\Sigma(R) = \Sigma_0 \left[1 + \left(\frac{R_b}{R} \right)^\alpha \right]^{\gamma/\alpha} \exp \left\{ -b_n \left[\frac{R^\alpha + R_b^\alpha}{R_e^\alpha} \right]^{1/(an)} \right\}, \quad (8)$$

where Σ_0 is the normalisation⁴, R_b is the break radius, R_e is the effective radius, n is the Sérsic index, γ is the logarithmic slope of the inner power law, and α controls the sharpness of the transition between the inner and outer regimes. The coefficient b_n is the standard Sérsic constant determined such that R_e encloses half of the total projected luminosity, i.e. b_n satisfies $\Gamma_i(2n, b_n) = \frac{1}{2}\Gamma_c(2n)$, where Γ_i and Γ_c denote the lower incomplete and complete gamma functions, respectively. All six quantities (Σ_0 , R_b , R_e , n , γ , α) are allowed to vary in the fit. In the following, R_b and γ provide our parametric estimates of the core radius and inner density slope, respectively.

The fits are performed using an MCMC sampling, with uniform priors

$$0 < \log_{10} \Sigma_0 < 25, \quad 0 < R_b < R_{\text{max}}, \quad 0 < R_e < R_{\text{max}}, \quad 0.1 < n < 10, \quad 0.1 < \alpha < \alpha_{\text{crit}}, \quad 0 < \gamma < 4, \quad (9)$$

where R_{max} corresponds to the maximum radial extent of the stellar distribution. The upper limit α_{crit} is a function of R_e , R_b , γ and n and corresponds to the maximum transition sharpness for which the deprojected core-Sérsic model remains physical (see Baes 2026, equations 5 and 6). We perform the fits over the radial range 0.02–100 kpc, where the lower limit corresponds to 6–10 times the softening length of the central particles. The upper limit is chosen to avoid extending the fit unnecessarily into the outer regions, where merger driven changes in the stellar envelope can affect the inferred Sérsic parameters. In particular, Hopkins & Hernquist (2010) showed that an extended outer envelope produced by a merger can modify the Sérsic fit and generate an apparent central mass deficit even when the inner density profile remains largely unchanged. The upper bound $n < 10$ is imposed in conjunction with this radial range to prevent pathological solutions in which the Sérsic index increases towards very large values in an attempt to reproduce an

⁴ The surface density Σ_b at the break radius can be evaluated from the expression: $\Sigma_b = \Sigma_0 2^{\gamma/\alpha} \exp[-b_n (2^{1/\alpha} R_b/R_e)^{1/n}]$.

approximately power law outer density profile. Such solutions can bias the inferred core parameters and may require multi-component modelling (see Section 4.4 of [Rantala et al. 2024](#)). Observationally, [Rusli et al. \(2013\)](#) identify unusually large Sérsic indices together with effective radii extending beyond the available radial range as signatures of this behaviour. In our fits, the posterior solutions remain below $n = 10$, indicating that this pathological regime is not encountered. Fiducial core-Sérsic parameters are obtained from the posterior distributions. For reference, the typical relative uncertainties associated with the MCMC fits are $\sim 1\%$ and $\sim 8\%$ for R_b and γ , respectively. The corresponding projection-induced scatter is slightly larger, at $\sim 4\%$ for R_b and $\sim 11\%$ for γ (see Section A for further details).

Non-parametric core properties. As an independent measure of the core properties, we also estimate the core radius and inner density slope without relying on the core-Sérsic model. We define the non-parametric core radius, R_γ , as the radius at which the logarithmic slope of the surface density profile reaches $-1/2$, $d \log \Sigma(R)/d \log R|_{R=R_\gamma} = -\frac{1}{2}$ ([Dullo & Graham 2012](#)).

The corresponding non-parametric inner slope is obtained by fitting a power law to the surface density profile at radii $R < R_\gamma/2$. Although this quantity is obtained from a one-parameter fit, we refer to it as the ‘non-parametric’ slope for simplicity and to distinguish it consistently from the slope γ obtained from the core-Sérsic fit.

Ejected mass. We estimate the stellar mass ejected by the SMBH binary by comparing the enclosed stellar mass in otherwise identical simulations with and without black holes. The inferred ejected mass depends on the radius at which this comparison is performed. At small radii, the enclosed mass difference may underestimate the total mass ejected by the binary, because stars initially located outside the chosen radius can be scattered to larger radii. Conversely, at sufficiently large radii, the mass difference may include stars that have been redistributed by the merger but remain gravitationally bound within the considered radius. We therefore define a characteristic ejected mass by identifying the maximum of the enclosed mass difference profile.

Specifically, we define

$$M_{\text{ej}}(r) = M_{\text{noBH}}(r) - M_{\text{BH}}(r), \quad (10)$$

where $M_{\text{noBH}}(r)$ and $M_{\text{BH}}(r)$ are the enclosed stellar masses at radius r in the simulations without and with black holes, respectively. Figure 2 shows seven representative profiles of the enclosed mass difference, expressed in units of the binary mass, $M_{\text{ej}}(r)/M_{\text{bin}}$, with the maximum of each profile marked by a black point. In most cases, this maximum is located close to the three-dimensional core radius, r_b . This radius is obtained from a core-Sérsic fit to the stellar density profile but in contrast to the projected fits described above, it is performed on three-dimensional ellipsoidal radial bins that account for the intrinsic shape of the remnant, providing a more accurate estimate of the radius marking the onset of the core.

For most remnants, the radius corresponding to the maximum of $M_{\text{ej}}(r)$ therefore coincides with r_b . For the purple cases shown in Fig. 2, this maximum lies within 6% of r_b , whereas in the green cases, where the maximum occurs at larger radii, it exceeds $1.5 r_b$ and reaches up to $\sim 2.5 r_b$ ⁵. We adopt the maximum value,

⁵ [Dosopoulou et al. \(2021\)](#) found that, in nonspherical merger remnants, the stars ejected by the SMBH binary can originate from radii larger than the core radius, in contrast to equivalent spherical models. However, they found

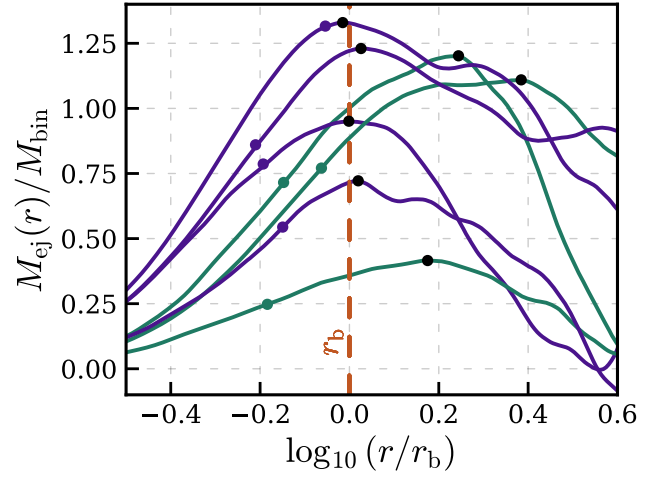


Figure 2. Radial profiles of the enclosed ejected mass, expressed in units of the binary mass, $M_{\text{ej}}(r)/M_{\text{bin}}$, for seven representative merger remnants. The black points mark the maximum of each profile, corresponding to our fiducial estimate $M_{\text{ej}}^{\text{max}}$. The vertical dashed line indicates the 3D core radius, r_b , obtained from a core-Sérsic fit to the stellar density profile using 3D ellipsoidal radial bins. Purple profiles are systems for which the maximum ejected mass is located close to r_b , while green profiles correspond to systems where the maximum occurs at larger radii. The coloured points indicate $M_{\text{ej}}(R_b)$, evaluated at the 2D core-Sérsic break radius R_b .

$M_{\text{ej}}^{\text{max}} \equiv \max_r M_{\text{ej}}(r)$, as our fiducial estimate of the ejected mass. For comparison, we also report $M_{\text{ej}}(R_b)$, evaluated at the two-dimensional core-Sérsic break radius.

Mass deficit. Observationally, the stellar mass deficit is inferred from the deficit in surface brightness relative to the inward extrapolation of the outer Sérsic profile, after adopting an appropriate stellar mass-to-light ratio ([Kormendy & Bender 2009](#); [Rusli et al. 2013](#); [Dullo & Graham 2014](#)). We follow the same approach for our simulated remnants, replacing the surface brightness profile with the stellar surface density profile. The mass deficit is therefore defined as the integrated difference between the best-fitting outer Sérsic profile, $\Sigma_{\text{Sérsic}}$, and the measured surface density profile, $\Sigma(R)$ ⁶,

$$M_{\text{def}} = 2\pi \int_0^\infty R dR [\Sigma_{\text{Sérsic}}(R) - \Sigma(R)]. \quad (11)$$

Here, $\Sigma_{\text{Sérsic}}$ is obtained by fitting a Sérsic profile to the outer regions of the remnant ($R > R_b$) and extrapolating it towards the centre.

2.3.4 Times selected for remnant analysis

We define two reference times to characterise the remnant before and after SMBH binary scouring. The pre-scouring remnant is evaluated

the energy distributions of the ejected stars to remain similar between the two geometries during the early hardening phase, from a_h to $a_h/3$, with the differences emerging at smaller binary separations. Since our simulations are analysed at $a_h/3$, this effect may therefore not explain the difference in the radius at which the maximum ejected mass occurs between the purple and green cases.

⁶ We have verified that restricting the integration to R_b , or to a few times R_b as adopted in some studies, does not affect the results presented below.

at t_b , the time at which the two SMBHs form a bound binary. Following François et al. (2026), this time is defined as the moment when the binding energy of the SMBH pair,

$$E_b = \frac{1}{2} \frac{M_1 M_2}{(M_1 + M_2)} v_{\text{rel}}^2 - \frac{G M_1 M_2}{d_{\text{rel}}}, \quad (12)$$

becomes negative and remains so for the rest of the simulation. With M_1 and M_2 the masses of the two SMBHs, and d_{rel} and v_{rel} their relative distance and relative velocity, respectively.

Defining a corresponding post-scouring time is less straightforward. Our simulations do not include post-Newtonian corrections and therefore do not model the final coalescence of the SMBHs. We therefore adopt the time at which the binary reaches a semi-major axis of $a = a_h/3$, where a_h is the hard binary separation, defined as

$$a_h = \frac{q_{\text{BH}}}{(1 + q_{\text{BH}})^2} \frac{r_{\text{inf}}}{4}, \quad (13)$$

with q_{BH} the black hole mass ratio and r_{inf} the influence radius i.e. the radius enclosing a stellar mass equal to twice the mass of the more massive SMBH. We refer to this epoch as the post-scouring time and use it as the common endpoint for the measurement of the remnant properties presented below.

For several mergers, we extended the simulations beyond $a_h/3$ to assess the impact of this choice on the inferred remnant properties. We find that the inner slope γ remains approximately unchanged after the binary reaches $a_h/3$, while the ejected stellar mass M_{ej} increases substantially until the binary reaches a_h , with only limited further evolution at smaller separations. In contrast, the core radius R_b continues to grow beyond $a_h/3$, in agreement with Dosopoulou et al. (2021), who found that it continues to increase down to approximately $a_h/5$. Figure B1 illustrates this behaviour by showing the evolution of R_b and γ for a subset of the mergers. The implications and limitations of adopting $a_h/3$ as the endpoint of our analysis are discussed in Section 4.3.

2.4 Scaling relations

2.4.1 Method for finding the scaling relations

A variety of statistical and machine learning techniques can be used to search for relations between galaxy properties, including regularised linear models such as lasso (Tibshirani 1996), tree based ensembles (Breiman 2001; Friedman 2001), Gaussian process emulators (Kimeldorf & Wahba 1970; Rasmussen & Williams 2006), nonlinear dimensionality reduction methods such as kernel PCA (Schölkopf et al. 1998), and topological data analysis (Chazal & Michel 2021). However, our training sample contains only 28 high-resolution simulations, making highly flexible models difficult to constrain reliably and less straightforward to interpret physically. We therefore adopt an empirical approach based on linear regression combined with best subset selection. This provides a transparent framework that is appropriate for our limited sample and produces compact scaling relations that are straightforward to interpret.

We first apply a logarithmic transformation to parameters spanning several orders of magnitude, such as masses and radii, to capture potential power law dependencies, while retaining parameters with more limited and physically bounded ranges, such as T and e , in their original form. The variables are then standardised by removing their respective means and dividing by their standard deviations. For a model containing n predictors selected from a total set of n_{tot} parameters, we perform an exhaustive search over all $\binom{n_{\text{tot}}}{n}$ possible predictor combinations. A linear regression is fitted for each combination, and its goodness of fit is quantified using the coefficient of

determination⁷, R^2 , defined as

$$R^2 = 1 - \frac{\sum_{i=1}^k (y_i - f_i)^2}{\sum_{i=1}^k (y_i - \bar{y})^2}, \quad (14)$$

where k is the number of data points, y_i denotes the measured value, f_i the corresponding model prediction, and $\bar{y}$ the sample mean.

We additionally impose a constraint on the degree of multicollinearity between predictors using the variance inflation factor (VIF). For a given predictor X_i , this quantity is defined as

$$\text{VIF}_i = \frac{1}{1 - R_i^2}, \quad (15)$$

where R_i^2 is obtained by regressing X_i against all other predictors included in the model. The VIF therefore quantifies the inflation of the uncertainty in a regression coefficient caused by linear correlations between predictors. For each candidate relation, we compute the VIF of every predictor and use the largest value, VIF_{max} , to characterise the overall multicollinearity of the model. We reject relations for which $\text{VIF}_{\text{max}} > 5$, thereby excluding models containing strongly correlated predictors.

Finally, we evaluate the predictive robustness of the candidate relations using leave-one-out cross-validation (LOOCV). Each data point is successively removed from the sample, the model is refitted to the remaining $k - 1$ systems, and the resulting model is used to predict the excluded point. The procedure is repeated for every system in the sample, and the resulting prediction errors are summarised through the root mean square error (RMSE). The final relation is selected by considering the combination of a high R^2 , a low VIF_{max} , and a low LOOCV RMSE.

When extending a relation from n to $n + 1$ predictors, we additionally assess whether the improvement in predictive power associated with the additional parameter is statistically supported. We do so using a partial F -test, which compares the reduction in the residual sum of squares (RSS) between the simpler and more complex models relative to the residual variance of the latter. The corresponding statistic is

$$F_{\text{partial}} = \frac{(\text{RSS}_r - \text{RSS}_c)/(p_c - p_r)}{\text{RSS}_c/(N - p_c)}, \quad (16)$$

where the subscripts r and c denote the reduced and complete models, respectively, p_r and p_c are their respective numbers of predictors, and N is the number of data points. Under the null hypothesis that the additional predictors do not provide genuine explanatory power, F_{partial} follows an F -distribution with $(p_c - p_r, N - p_c)$ degrees of freedom. The associated p -value therefore quantifies the probability of obtaining an improvement at least as large as that observed purely from statistical fluctuations. We require this p -value to be below 0.05 when assessing whether the addition of a predictor provides a statistically supported improvement over the simpler relation.

2.4.2 Parameters considered

For each scaling relation analysis, we explore the dependence of the target quantity on the following set of remnant, progenitor, and orbital parameters:

⁷ The coefficient of determination measures the fraction of the variance in the dependent variable explained by the regression model, ranging from 0 for no explanatory power to 1 for a perfect fit.

■ **Scale lengths.** We consider the stellar half-mass radius of the remnant, r_{hm} , and the influence radius of the SMBH binary in the remnant at binary formation, r_{inf} .

■ **Masses.** We include the dark matter, stellar, and total masses of the first and second progenitors, denoted by $M_{\text{DM,FP}}$, $M_{\text{DM,SP}}$, $M_{\text{★,FP}}$, $M_{\text{★,SP}}$, $M_{\text{tot,FP}}$, and $M_{\text{tot,SP}}$, respectively. The SMBH masses, $M_{\text{BH,FP}}$ and $M_{\text{BH,SP}}$, and the corresponding mass ratios, q_{DM} , $q_{\text{★}}$, q_{tot} , and q_{BH} are also considered, with all mass ratios defined as the mass of the first progenitor divided by that of the second progenitor. We define the binary mass as $M_{\text{bin}} = M_{\text{BH,FP}} + M_{\text{BH,SP}}$. We further consider the initial stellar mass enclosed within r_{inf} and $5r_{\text{inf}}$ for each progenitor, denoted by $M_{\text{enc,FP}}(r_{\text{inf}})$, $M_{\text{enc,SP}}(r_{\text{inf}})$, $M_{\text{enc,FP}}(5r_{\text{inf}})$, and $M_{\text{enc,SP}}(5r_{\text{inf}})$. For the remnant, considering particles originating from both progenitors at $t = t_{\text{b}}$, we similarly compute $M_{\text{enc,rem}}(r_{\text{inf}})$ and $M_{\text{enc,rem}}(5r_{\text{inf}})$.

■ **Densities.** We evaluate the stellar density of each progenitor at r_{inf} , $5r_{\text{inf}}$, and r_{hm} , i.e. $\rho_{\text{★,FP}}$ and $\rho_{\text{★,SP}}$ evaluated at these radii. These quantities are measured both at the beginning of the simulation and at the time of binary formation, $t = t_{\text{b}}$. For the remnant, using the combined stellar population of the two progenitors at $t = t_{\text{b}}$, we similarly evaluate the density at these three characteristic radii, denoted by $\rho_{\text{★,rem}}(r_{\text{inf}})$, $\rho_{\text{★,rem}}(5r_{\text{inf}})$, and $\rho_{\text{★,rem}}(r_{\text{hm}})$.

■ **Orbit.** For the galaxy encounter, we consider the initial semi-major axis a_0 , eccentricity e_0 , pericentre r_{peri} , and apocentre r_{apo} . We also define the interpenetration parameter $r_{\text{inter}} = r_{\text{hm}}/r_{\text{peri}}$. For the SMBH binary, we consider the semi-major axis, eccentricity, and pericentre at binary formation, denoted by a_{b} , e_{b} , and $r_{\text{peri,b}}$, respectively.

■ **Shape & Kinematics.** The intrinsic shape of the remnant is characterised by the triaxiality parameter T . We characterise the stellar kinematics using the rotational support v_{ϕ}/σ_{ϕ} , the one-dimensional velocity dispersion σ , and the velocity anisotropy parameter β , evaluated at r_{inf} , $5r_{\text{inf}}$, and r_{hm} . We also include β_{max} , defined as the maximum value of the anisotropy profile.

■ **Core properties.** We consider the core-Sérsic break radius R_{b} , effective radius R_{e} , inner slope γ , and Sérsic index n , together with the non-parametric core radius R_{γ} and inner slope. We additionally include the maximum ejected stellar mass, $M_{\text{ej}}^{\text{max}}$, and the stellar mass deficit, M_{def} .

All distances are expressed in kpc, masses in $M_{\odot}$, densities in $M_{\odot} \text{ kpc}^{-3}$, and velocities in km s^{-1} .

3 RESULTS

3.1 Shape and kinematics

3.1.1 Shape

Figure 3 shows the probability density function of the remnant shapes in our N -body simulations, evaluated at $a = a_{\text{h}}/3$, for runs performed with and without central black holes. We first consider the central shape, measured within the SMBH binary influence radius, r_{inf} , shown by the dotted curves in the left-hand panel. A clear difference is found between the simulations with and without SMBHs. The remnants with SMBHs preferentially have lower triaxiality, with $T \lesssim 0.5$, whereas those without SMBHs are predominantly prolate, with $T \gtrsim 0.5$. These values should nevertheless be interpreted with some caution. The central regions are close to spherical, with both axis ratios close to unity (see Fig. C1), making $T = (1 - p^2)/(1 - q^2)$ highly sensitive to small changes in p and q . This naturally produces

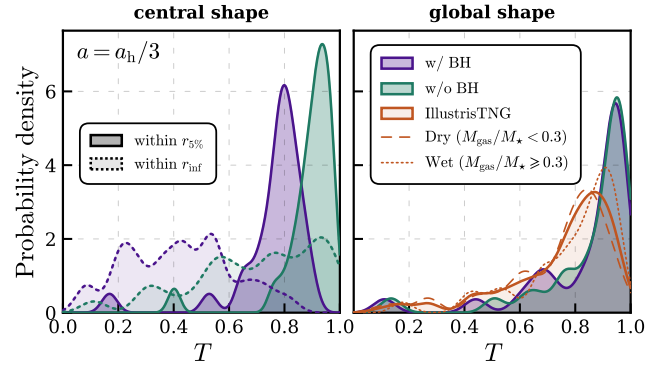


Figure 3. Probability density function of the stellar triaxiality parameter T measured at $a = a_{\text{h}}/3$ for the merger remnants in our N -body simulations. The left panel shows the central shape, measured within the binary influence radius r_{inf} (dotted curves) and within the radius enclosing 5% of the total stellar mass (solid curves), while the right panel shows the global shape, measured from the 50% most bound stellar particles. The purple and green curves correspond to simulations performed with and without central black holes, respectively. The orange curve shows the distribution for the same systems in ILLUSTRISTNG. For reference, the dashed and dotted orange lines indicate the distributions for dry ($M_{\text{gas}}/M_{\text{★}} < 0.3$) and wet ($M_{\text{gas}}/M_{\text{★}} \geq 0.3$) subsamples in ILLUSTRISTNG. The presence of a central black hole binary affects the central shape, with remnants hosting black holes having systematically lower values of T than their counterparts without SMBHs. On galactic scales, the remnants are predominantly prolate, with distributions broadly consistent with ILLUSTRISTNG, although cosmological systems are slightly more triaxial, likely reflecting environmental effects. The distributions peak at $T = 0.95$ for the N -body remnants and at $T = 0.87$ for the ILLUSTRISTNG galaxies.

broad distributions of T even for small departures from spherical symmetry. The systematic shift towards lower T in the simulations with SMBHs is, however, a genuine feature that persists at larger radii. The solid curves show the shape measured within an intermediate radius, $r_{5\%}$, defined as the radius enclosing 5% of the total stellar mass. At this scale, the remnants without SMBHs have a triaxiality distribution closely resembling their global shape, with a similar peak at $T \approx 0.95$, whereas the remnants with SMBHs remain shifted towards lower T values. This indicates that the presence of the SMBH binary tends to make the remnant less prolate in its inner regions.

The right-hand panel of Fig. 3 shows the global remnant shapes. For comparison, we also include the distribution of shapes for the same galaxies extracted from ILLUSTRISTNG. In this latter case, no selection based on gas fraction and black hole mass is applied, although we separately indicate the distributions corresponding to dry ($M_{\text{gas}}/M_{\text{★}} < 0.3$) and wet ($M_{\text{gas}}/M_{\text{★}} \geq 0.3$) systems. We find that the merger remnants in our simulations are predominantly prolate, with a strong preference for high triaxiality values, $T \gtrsim 0.8$. The distributions obtained from simulations with and without black holes are very similar, indicating that the presence of central black holes has a negligible impact on the global shape of the remnant at this stage of the evolution. The corresponding distributions of the intrinsic axis ratios, b/a and c/a , are shown in Appendix C for readers interested in the detailed shape properties.

In comparison, the ILLUSTRISTNG galaxies are slightly less prolate, with a peak of the distribution at $T = 0.87$, compared to $T = 0.95$ for the remnants of our N -body simulations (both with and without black holes). The ILLUSTRISTNG distribution is also broader. This difference likely reflects the idealised nature of our initial conditions:

our progenitor galaxies are initially spherical, whereas the shapes of galaxies in cosmological simulations are determined by their assembly histories and environments, including previous mergers. As a result, cosmological galaxies tend to display a wider range of intrinsic shapes and a tendency towards more triaxial configurations than those obtained in our controlled merger experiments.

Figure 4 illustrates the projected surface density of a representative merger remnant, selected to have a triaxiality parameter close to the peak of the distribution shown in Fig. 3. The left and right panels show projections in the (x, y) and (x, z) planes, respectively. Overlaid black contours indicate isodensity levels. The remnant exhibits a clear elongation along one principal axis, characteristic of a prolate morphology.

To investigate the origin of the predominantly prolate remnants, we examine the connection between the remnant shape and the orbital properties of the galaxy merger. The left-hand panel of Fig. 5 shows the triaxiality parameter as a function of the initial orbital eccentricity. We find a clear correlation between these quantities, with more eccentric mergers preferentially producing more prolate remnants. This result indicates that the high fraction of prolate systems identified in our sample is primarily driven by the strongly radial nature of cosmological mergers (see François et al. 2026).

We quantify this trend by deriving a scaling relation between the triaxiality parameter and the orbital eccentricity (relation a1), reported in Table 2. This relation achieves a coefficient of determination $R^2 = 0.87$, already indicating strong predictive power for a single-parameter model.

We further explore whether the inclusion of an additional parameter can improve the predictive accuracy. We find that the semi-major axis of the galactic encounter provides the most significant improvement when combined with the eccentricity. The resulting two-parameter relation (equation a2) yields a higher coefficient of determination, $R^2 = 0.90$.

We also find that the remnant shape is strongly linked to its rotational support. The middle panel of Fig. 5 shows the triaxiality parameter as a function of the rotation parameter v_ϕ/σ_ϕ , measured within the half-mass radius. A tight correlation is observed, with systems characterised by higher triaxiality exhibiting lower rotational support. The scaling relation between shape and rotation is given in equation (a3) of Table 2. It achieves a very high coefficient of determination of $R^2 = 0.96$, highlighting the strength of the connection between galaxy shape and kinematics.

3.1.2 Rotation

The connection between shape and rotation can be further understood by linking both quantities to the orbital properties of the merger. The right-hand panel of Fig. 5 shows the rotation parameter v_ϕ/σ_ϕ , measured within the half-mass radius, as a function of the initial orbital eccentricity. A clear correlation is observed, with more eccentric mergers producing remnants with lower rotational support. The corresponding scaling relation is given in equation (b1) of Table 2, with a coefficient of determination of $R^2 = 0.89$.

This behaviour is fully consistent with the trends identified above: mergers on more radial orbits lead to remnants that are both more prolate and more dispersion supported. We further improve the predictive power by introducing a second parameter, namely the semi-major axis of the encounter. The resulting two-parameter relation, given in equation (b2) of Table 2, reaches $R^2 = 0.91$.

Overall, these results demonstrate that both the shape and the rotational support of the remnant are primarily set by the initial

orbital configuration of the merger, with tight and interconnected relations linking triaxiality, rotation, and eccentricity.

The first two columns of Fig. 6 show the probability density function of v_ϕ/σ_ϕ in our N -body simulations, measured both before binary hardening ($t = t_b$) and at $a = a_h/3$. The rotation parameter v_ϕ/σ_ϕ is evaluated within the binary influence radius and within the stellar half-mass radius, for simulations performed with and without central black holes. For comparison, we also include the corresponding distributions for the same galaxies extracted from ILLUSTRISTNG.

As discussed above, the predominantly high eccentricities of the mergers imply that only a small amount of angular momentum is transferred to the remnant. As a result, the systems are generally slow rotators, with $v_\phi/\sigma_\phi \lesssim 0.3$ within the half-mass radius in most cases. We find no significant difference in the rotational support between simulations with and without black holes, both on galactic scales (within the half-mass radius) and in the central regions (within the binary influence radius).

The rotation distribution of ILLUSTRISTNG galaxies is remarkably similar to that of our remnants. Although cosmological galaxies exhibit slightly higher rotation, with a peak at $v_\phi/\sigma_\phi = 0.06$ compared to $v_\phi/\sigma_\phi = 0.03$ in our simulations, both distributions are consistent with systems that are predominantly dispersion supported.

3.1.3 Anisotropy

The last two columns of Fig. 6 show the probability density function of the anisotropy parameter in our N -body simulations, measured both before binary hardening ($t = t_b$) and at $a = a_h/3$. The parameter is evaluated within the binary influence radius and within the stellar half-mass radius, for simulations performed with and without central black holes. For comparison, we also include the corresponding distribution within the half-mass radius for the same galaxies extracted from ILLUSTRISTNG.

Within the binary influence radius, the simulations with and without black holes already exhibit slight differences in their anisotropy distributions before binary hardening. Nevertheless, both distributions remain confined to a similar range of values, with $\beta \approx 0$ –0.1, indicating mildly radially anisotropic stellar orbits. Following the scouring phase, however, the difference becomes much more pronounced. Simulations including black holes exhibit a clear shift towards lower β values, with a substantial fraction of remnants reaching $\beta < 0$, corresponding to tangential anisotropy. In contrast, simulations without black holes remain mildly radially anisotropic, with β still confined to the range ~ 0 –0.1. This transition to tangential anisotropy is a direct consequence of binary scouring: the SMBH binary preferentially ejects stars with low angular momentum through three-body interactions, thereby depleting the radial orbit population and leaving the central stellar distribution dominated by tangential orbits.

At the scale of the stellar half-mass radius, no significant differences are found between simulations with and without black holes. In both cases, the remnants remain radially anisotropic, with typical values of $\beta \approx 0.35$ –0.55.

A comparison with ILLUSTRISTNG reveals that the cosmological galaxies are systematically less radially anisotropic, with their distributions peaking around $\beta \approx 0.3$ –0.4. Part of this offset can be attributed to the presence of gas, as illustrated by the difference between the wet and dry ILLUSTRISTNG subsamples. As shown in Fig. D1, progressively selecting galaxies with lower gas fractions shifts the ILLUSTRISTNG distribution towards higher β values. Indeed, gas dissipation during mergers promotes the formation of short axis tube (z-tube) orbits at the expense of radially biased box orbits, driving β

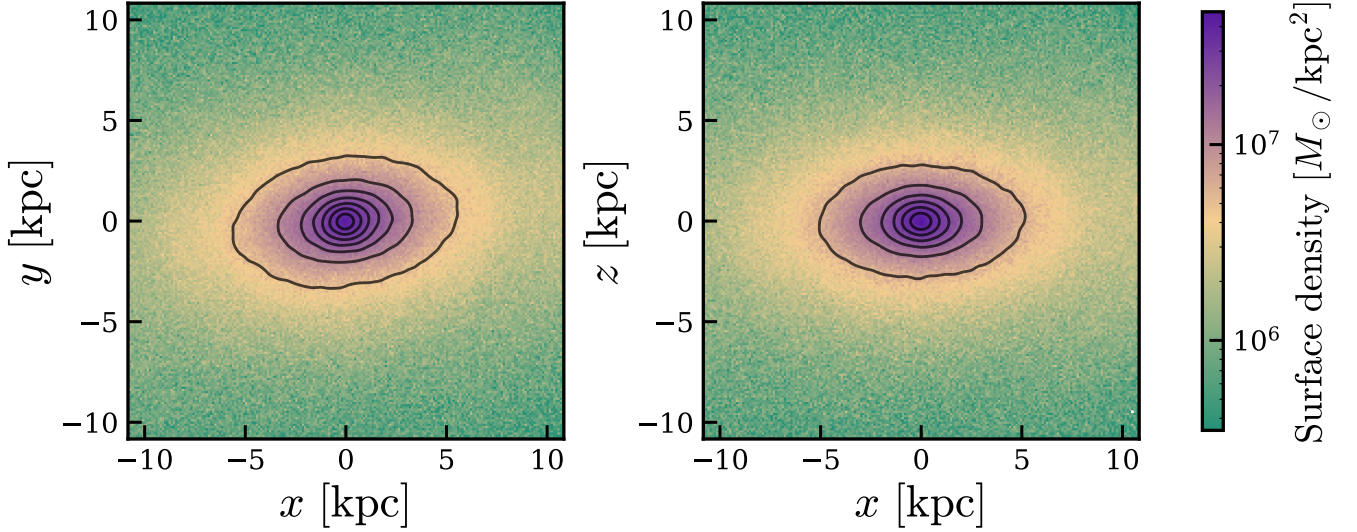


Figure 4. Projected surface density maps of a representative merger remnant with triaxiality parameter close to the peak of the distribution ($T = 0.95$). The left and right panels show projections in the (x, y) and (x, z) planes, respectively. Black contours indicate isodensity levels. The remnant displays a clear elongation along a preferred axis, consistent with a predominantly prolate morphology.

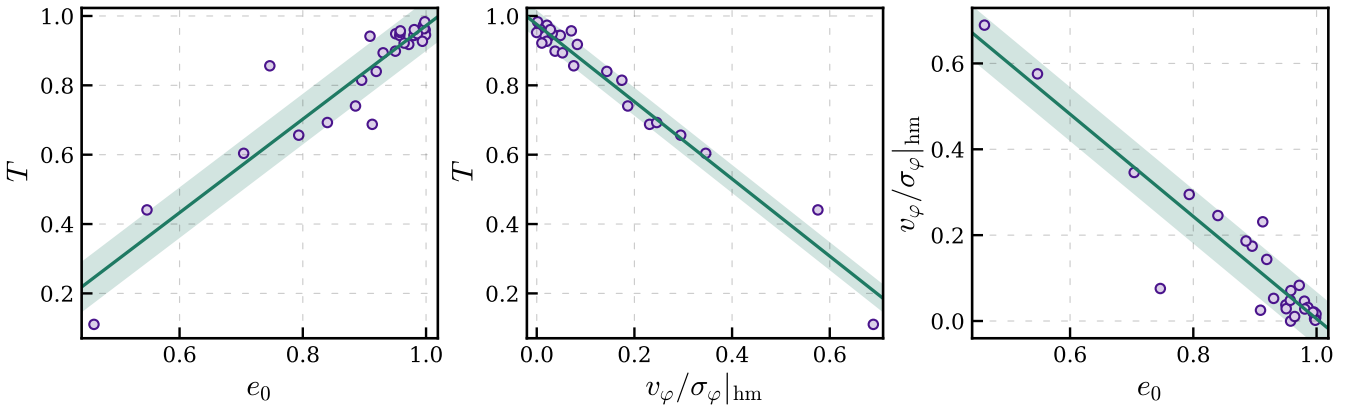


Figure 5. Correlations between the remnant shape, rotational support, and the initial orbital eccentricity of the merger. The left panel shows the triaxiality parameter, T , as a function of the initial orbital eccentricity, e_0 . The middle panel shows T as a function of the stellar rotation parameter, v_ϕ/σ_ϕ , measured within the half-mass radius. The right panel shows v_ϕ/σ_ϕ as a function of e_0 . Solid lines indicate the best fitting linear relations, with the shaded regions representing the intrinsic scatter of the relation. More eccentric mergers produce increasingly prolate remnants with lower rotational support.

toward more tangential or isotropic values (Cox et al. 2006; Hoffman et al. 2010). Nevertheless, even in the limit of vanishing gas fraction ($M_{\text{gas}}/M_\star \rightarrow 0$), corresponding most closely to our gas free simulations, the ILLUSTRITNG distribution remains systematically offset from our remnants. This suggests that gas content alone cannot fully account for the observed difference in velocity anisotropy. The remaining discrepancy likely reflects differences between our idealized merger models and fully cosmological galaxies, including differences in their initial orbital structure and assembly histories. Disentangling the relative importance of these effects, however, requires a dedicated analysis and is left for future work.

To further investigate the origin of the radial anisotropy measured within the stellar half-mass radius, Fig. 7 presents the radial profiles of the velocity anisotropy parameter for the 28 merger remnants, colour coded according to the initial orbital eccentricity of the merger.

The anisotropy profiles exhibit a clear dependence on the merger eccentricity. Mergers with $e_0 \gtrsim 0.6$ develop a pronounced bell-shaped profile, with the radial anisotropy reaching its maximum at $r \simeq 0.1 d_{\text{ini}}$, where d_{ini} is the initial separation between the two progenitors (equal to the apocentric distance in our simulations). In contrast, remnants produced by less eccentric mergers ($e_0 \lesssim 0.6$) display a weaker radial anisotropy around the same characteristic radius, accompanied by a tangentially biased region extending over approximately $-0.5 \lesssim \log_{10}(r/d_{\text{ini}}) \lesssim 0$.

The systematic dependence of the anisotropy profile on the merger eccentricity indicates that progressively more radial encounters produce remnants with increasingly radially biased stellar orbits. This trend is further illustrated in Fig. 8, which shows a clear positive correlation between the maximum value of the anisotropy parameter, β_{max} , and the initial orbital eccentricity.

Table 2. Scaling relations for the global remnant shape, rotation, and velocity anisotropy. Relations a1–a3 model the remnant triaxiality T , relations b1–b2 model the rotational support $(v_\phi/\sigma_\phi)_{\text{hm}}$ and relations c1–c4 model the maximum stellar velocity anisotropy β_{max} . For each relation we report: the number of predictors n_{var} ; the coefficient of determination R^2 ; the adjusted coefficient of determination R^2_{adj} , which penalises model complexity and allows fair comparison across relations with different numbers of predictors; the intrinsic scatter Δ ; the maximum variance inflation factor (VIF), which quantifies multicollinearity among predictors; and the root mean square error of a leave-one-out cross validation (LOOCV), which estimates out of sample predictive performance.

Relation		n_{var}	R^2	R^2_{adj}	Δ	VIF	LOOCV
Shape							
a1	$T = +1.35 e_0 - 0.38$	1	0.87	0.87	0.07	✗	0.43
a2	$T = +1.22 e_0 + 0.11 \log_{10}(a_0/\text{kpc}) - 0.44$	2	0.90	0.89	0.06	1.3	0.39
a3	$T = -1.11 (v_\phi/\sigma_\phi)_{\text{hm}} + 0.98$	1	0.96	0.95	0.04	✗	0.39
Rotation							
b1	$(v_\phi/\sigma_\phi)_{\text{hm}} = -1.20 e_0 + 1.21$	1	0.89	0.88	0.06	✗	0.43
b2	$(v_\phi/\sigma_\phi)_{\text{hm}} = -1.08 e_0 - 0.10 \log_{10}(a_0/\text{kpc}) + 1.27$	2	0.91	0.91	0.05	1.3	0.39
Anisotropy							
c1	$\beta_{\text{max}} = +0.76 e_0 - 0.23$	1	0.75	0.74	0.06	✗	0.53
c2	$\beta_{\text{max}} = +0.58 e_0 + 0.16 \log_{10}(a_0/\text{kpc}) - 0.32$	2	0.89	0.88	0.04	1.3	0.49
c3	$\beta_{\text{max}} = +0.55 T - 0.01$	1	0.82	0.82	0.05	✗	0.44
c4	$\beta_{\text{max}} = -0.61 (v_\phi/\sigma_\phi)_{\text{hm}} + 0.53$	1	0.79	0.78	0.05	✗	0.49

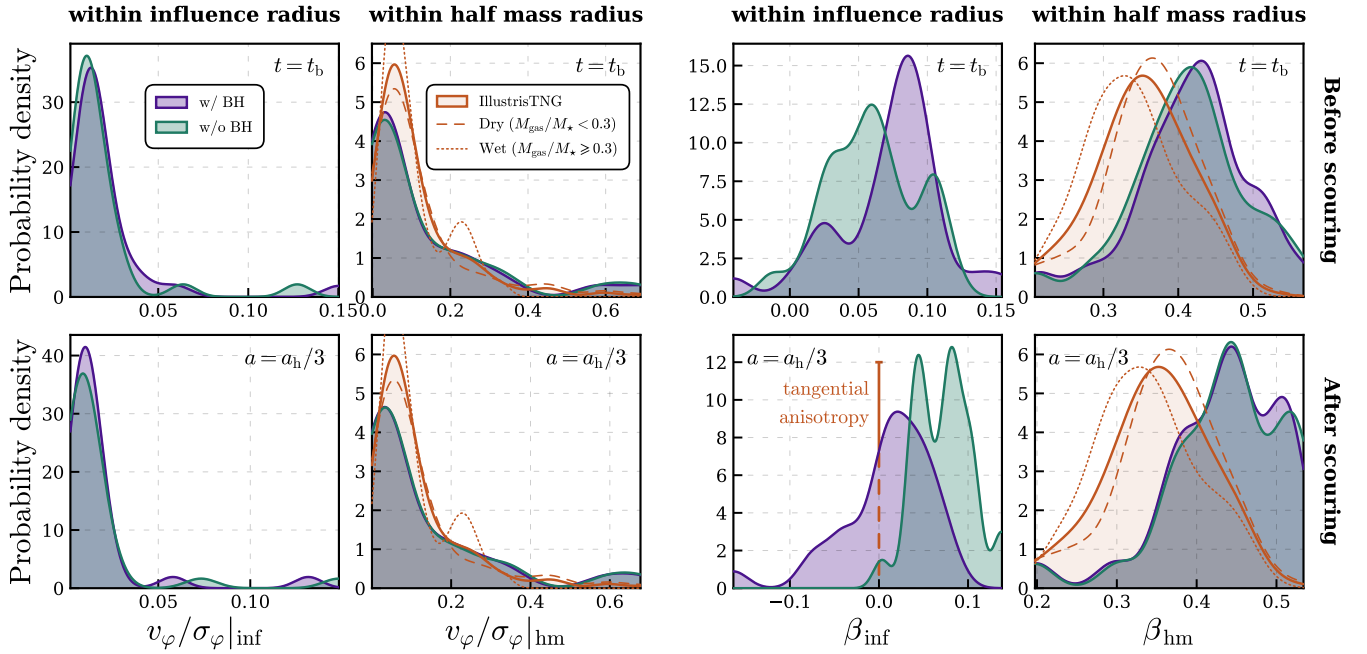


Figure 6. Probability density distributions of the stellar rotational support and velocity anisotropy. The first two columns show the stellar rotation parameter v_ϕ/σ_ϕ , while the last two show the velocity anisotropy parameter β . Within each pair, the left and right panels correspond to measurements within the binary influence radius and the stellar half mass radius, respectively. The upper panels show the distributions at the time of binary formation, $t = t_b$, while the lower panels correspond to the hardening stage at $a = a_h/3$. Green and purple curves show the distributions obtained from simulations without and with central black holes, respectively. For the half mass radius measurements, the orange curves show the corresponding distributions for galaxies in ILLUSTRISTNG, with dashed and dotted lines indicating the dry ($M_{\text{gas}}/M_\star < 0.3$) and wet ($M_{\text{gas}}/M_\star \geq 0.3$) merger subsamples, respectively. The remnants are predominantly slow rotators, with their rotational support largely unaffected by the presence of central black holes and broadly consistent with that of their ILLUSTRISTNG counterparts. In contrast, binary hardening drives the central stellar population towards tangential anisotropy, while the anisotropy measured within the half mass radius remains largely unaffected by the presence of black holes. Compared to our remnants, ILLUSTRISTNG galaxies are systematically less radially anisotropic on galactic scales.

The relation (c1) quantifies the dependence of β_{max} on the initial encounter eccentricity, yielding a high coefficient of determination of $R^2 = 0.75$. Including the semi-major axis of the galaxy encounter, a_0 , further improves the relation to $R^2 = 0.89$ (relation c2). As found for the remnant shape and rotational support, the anisotropy is

therefore strongly determined by the orbital properties of the galaxy encounter that forms the remnant. Since both the shape and rotation are themselves linked to the encounter orbit, it is also natural to find correlations between β_{max} and these structural properties. Indeed, we obtain single-parameter relations with the triaxiality parameter, T

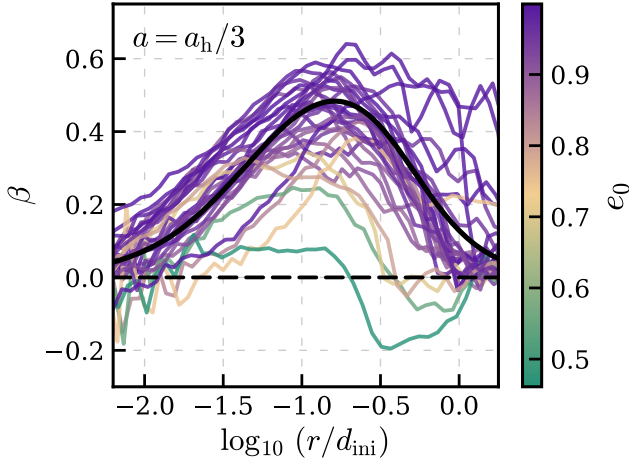


Figure 7. Radial profiles of the stellar velocity anisotropy parameter, β for the 28 merger remnants. Profiles are colour coded according to the initial orbital eccentricity of the merger, e_0 . More eccentric mergers develop stronger radial anisotropy, producing a pronounced bell-shaped profile that peaks at $r \approx 0.1 d_{\text{ini}}$, where d_{ini} is the initial separation between the progenitor galaxies. In contrast, remnants formed through less eccentric mergers exhibit weaker radial anisotropy and develop an extended tangentially biased region at intermediate radii. The black line shows an MCMC fit of the β profile corresponding to the most probable eccentricity of cosmological encounters, using a double sigmoid function.

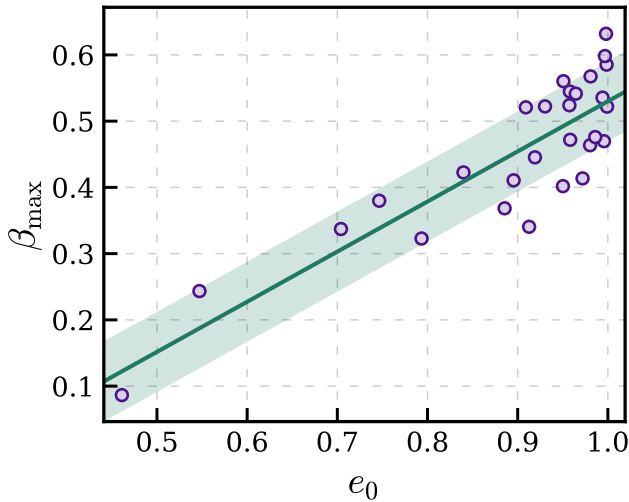


Figure 8. Maximum stellar velocity anisotropy, β_{max} , as a function of the initial orbital eccentricity, e_0 , for the 28 merger remnants. The solid line shows the best fitting linear relation, while the shaded region denotes the intrinsic scatter of the relation. More eccentric mergers systematically produce remnants with higher radial anisotropy peak.

(relation c3), and the rotational support, v_ϕ/σ_ϕ (relation c4), yielding $R^2 = 0.82$ and 0.79 , respectively.

We find that the β profile corresponding to the peak of the cosmological encounter eccentricity distribution, $e_0 = 0.94$ (see François et al. (2026)), is well captured by a double sigmoid function, whose parameters are determined through an MCMC fit (black line in

Fig. 7):

$$\beta(x) = A \left[\frac{1}{1 + \exp[-k_1(x - x_1)]} \right] \left[\frac{1}{1 + \exp[k_2(x - x_2)]} \right], \quad (17)$$

where

$$\begin{aligned} A &= +0.66 \pm 0.03, & k_1 &= +3.04 \pm 0.09, \\ k_2 &= +4.28 \pm 0.14, & x_1 &= -1.31 \pm 0.03, \\ x_2 &= -0.33 \pm 0.02, \end{aligned} \quad (18)$$

and $x = \log_{10}(r/d_{\text{ini}})$.

3.2 Core properties

Figure 9 shows six representative examples of the central stellar surface density profiles of the merger remnants at $a = a_h/3$, shown for one of the projections considered, for simulations with and without black holes. For the simulations including black holes, the best fitting core-Sérsic profiles obtained from our MCMC analysis are shown by the solid purple curves. The outer Sérsic fits, performed using the data points beyond R_b , are shown by the orange curves. The corresponding values of the core-Sérsic break radius, R_b , and the non-parametric core radius, R_γ , are indicated in each panel and the lower radial limit of 20 pc corresponds to 6–10 times the softening length of the central particles.

The surface density profiles of the remnants with and without black holes are remarkably similar. In particular, the inner slopes show no substantial systematic difference between the two sets of simulations, in contrast to the behaviour commonly found for more idealised initial conditions. The similarity of the inner slopes, including the relatively shallow values found in the simulations without black holes, can be understood from the relatively shallow inner density profiles of our progenitors (see Fig. E1). Their three-dimensional inner slopes $\gamma_{\text{pr},3\text{D}}$ are generally below $\gamma_{\text{pr},3\text{D}} = 1$, with the majority having $\gamma_{\text{pr},3\text{D}} \lesssim 0.7$. In projection, these correspond to inner slopes $\gamma_{\text{pr},2\text{D}}$ below ~ 0.3 , consistent with the inner slopes measured in the remnants of the simulations without black holes, which, as discussed in Section 3.2.1, range from $\gamma \approx 0.05$ to 0.25 . The modest further flattening of the remnant profiles compared to progenitors appears to result primarily from merger induced central heating, as also found in the idealised merger simulations of Bortolas et al. (2018) and Dosopoulou et al. (2021).

3.2.1 Core radius and inner slope

We compare the central structural properties of the merger remnants in simulations with and without black holes, measured at $a = a_h/3$.

Core radius. The left-hand panel of Fig. 10 shows the probability density distributions of the median core radius values measured over the 12 projections considered, characterised in two ways. The first is the break radius R_b obtained from a core-Sérsic fit to the density profile, while the second, R_γ , is defined non-parametrically as the radius at which the logarithmic density slope reaches $d \log \Sigma(R)/d \log R|_{R=R_\gamma} = -1/2$. R_b spans 0.2–2.0 kpc while R_γ spans 0.2–1.1 kpc. In both cases, the distributions are shifted towards smaller core radii in the absence of black holes, indicating that the presence of a central black hole binary is associated with the formation of larger cores. The values of R_b for each merger remnant, for simulations with and without black holes, are listed in Table A1.

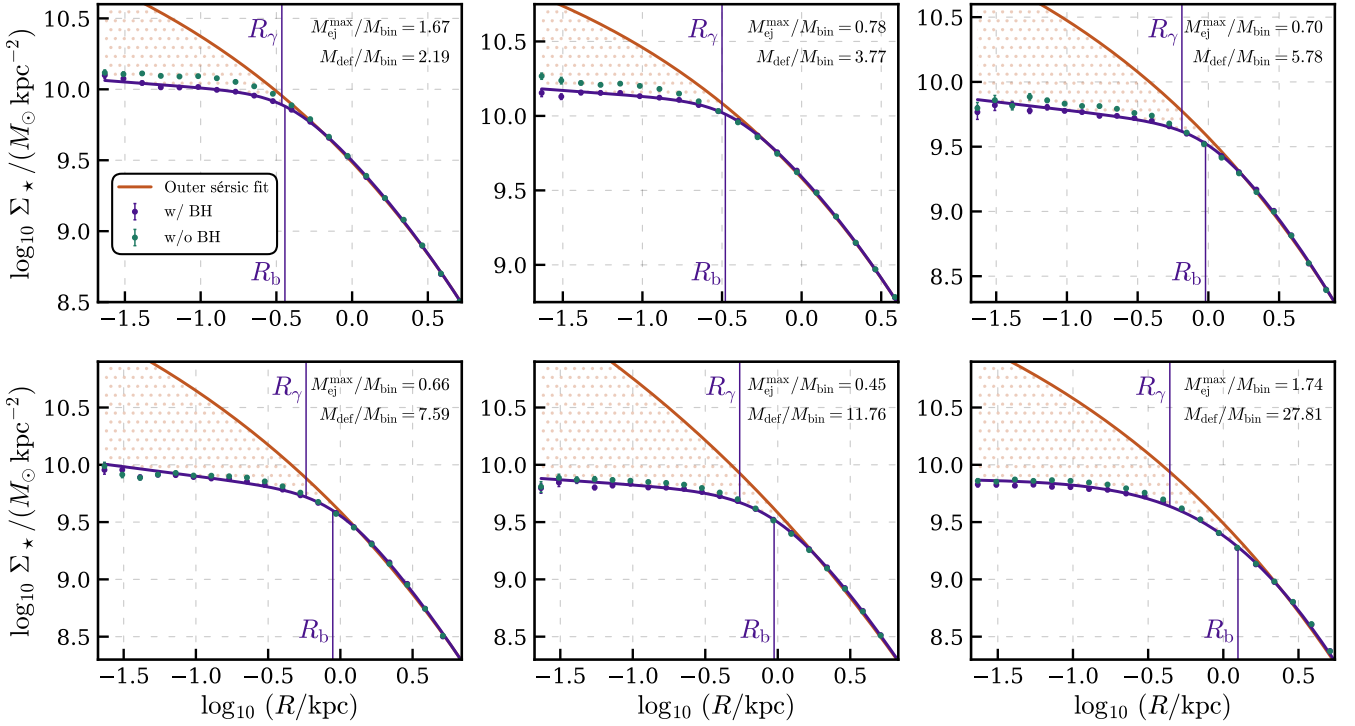


Figure 9. Stellar surface density profiles of six representative merger remnants at $a = a_h/3$, shown for one of the projections considered, for simulations with (purple) and without (green) black holes. The solid curves show the core-Sérsic fits obtained from our MCMC analysis for the simulations with black holes. The orange curves show Sérsic fits to the outer profiles, using only points at $R > R_b$. The vertical purple lines indicate the core-Sérsic break radius R_b and the non-parametric core radius R_{γ} , defined as the radius at which the logarithmic density slope reaches $-1/2$. The ratios of the maximum ejected mass, $M_{\text{ej}}^{\text{max}}$, and mass deficit, M_{def} , to the binary mass M_{bin} are reported in each panel. Despite the different black hole content, the inner surface density profiles remain broadly similar, with comparable core structures and inner slopes, a direct consequence of the relatively shallow inner density profiles of our ILLUSTRITNG-based progenitors.

Inner slope. The right-hand panel of Fig. 10 shows the corresponding distributions of the median inner logarithmic slope γ measured over the 12 projections considered. We again consider two estimates: the value obtained from the core-Sérsic fit and a ‘non-parametric’ (in reality one-parametric) estimate obtained by fitting a power law to the density profile within $R_{\gamma}/2$. The core-Sérsic values span approximately 0–0.27, while the non-parametric estimates are typically in the range 0.05–0.15. The core-Sérsic γ is particularly useful for comparison with observations and previous studies, where the inner slope is generally derived from core-Sérsic fits but because the core-Sérsic model is fitted over a broad radial range and must simultaneously describe the central profile and the outer Sérsic component, the resulting γ can retain some covariance with the other fitted parameters. Since the profile is better sampled at larger radii, these regions provide stronger constraints on the overall fit, leaving greater freedom in the inner region (see Section A for further details on the uncertainties). The non-parametric estimate is instead determined solely from the innermost points, allowing the central slope to be constrained with less influence from the outer profile. This comes at the cost of a less direct comparison with observational measurements. The apparent shift towards lower γ values in the non-parametric distributions without black holes is, however, an artefact of evaluating the slope within $R_{\gamma}/2$, as R_{γ} is smaller in the absence of black holes, causing the measured slope to appear systematically shallower. To isolate the effect of the black hole, we instead evaluate the slope over the same radial range, $R_{\gamma}/2$ defined from the simulations with black holes. The resulting distribution, shown in orange, shifts towards larger γ

values in the simulations without black holes, indicating that the inner density profiles are, on average, steeper in the absence of black holes.

Taken together, these results from Fig. 9 and 10 suggest that the shallow inner slopes of the merger remnants are largely inherited from the relatively shallow central profiles of the progenitors, with an additional flattening induced by merger driven central heating. The subsequent evolution of the SMBH binary through stellar scouring primarily affects systems with projected inner slopes $\gamma \gtrsim 0.15$, lowering them into the range $\gamma \approx 0.05$ –0.15, as indicated by the difference between the non-parametric slope distributions with and without black holes in Fig. 10. At still smaller slopes, the main effect of continued binary hardening is instead to extend the depleted region to larger radii, resulting in a progressive increase in the core radius.

Relations for the core radius. We next investigate scaling relations for the core radius, characterised by both R_b and R_{γ} , and for the inner density slope. Equation (d1) in Table 3 provides a single-predictor relation between R_{γ} and the SMBH binary influence radius, r_{inf} , yielding a remarkably high coefficient of determination of $R^2 = 0.93$. Among the parameters considered, r_{inf} provides the strongest correlation with the core radius, with larger influence radii corresponding to larger core radii, as expected. Since several studies have also linked core sizes to the final black hole mass, we additionally consider a relation based solely on M_{bin} , given by equation (d2), which yields a substantially lower $R^2 = 0.44$. This indicates that the

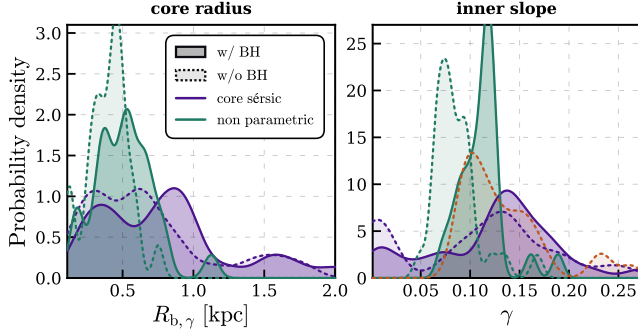


Figure 10. Probability density distributions of the core parameters of the merger remnants, measured at $a = a_h/3$, for simulations with (solid lines) and without (dotted lines) black holes. The distributions are constructed from the median values measured over the 12 projections considered. The left-hand panel shows the core radius measured using a core-Sérsic fit, R_b (purple), and the non-parametric core radius, R_γ (green), defined as the radius at which the logarithmic surface density slope reaches $-1/2$. The right-hand panel shows the corresponding inner slope γ , measured either from the core-Sérsic fit (purple) or by fitting the density profile within $R_\gamma/2$ (green). The orange distribution shows the corrected version of the dotted green distribution, with the inner slope evaluated over the same radial range as in the simulations with black holes, using $R_\gamma/2$ measured from the black hole simulations. The black hole runs produce systematically larger core radii, while the comparison of the inner slopes indicates that the central density profiles are, on average, steeper in the absence of black holes when evaluated over a common radial range.

core radius retains information about the binary mass, but that r_{inf} provides a more informative predictor by simultaneously encoding the binary mass and the central mass distribution of the host galaxy.

Equation (d3) extends this relation to two predictors, combining r_{inf} with the binary pericentre at formation. This yields $R^2 = 0.95$, indicating that the initial binary orbit provides an additional, albeit modest, contribution to setting the core radius. Although the improvement over the single-predictor relation is relatively small for R_γ , it is statistically significant, and the gain in predictive power is more pronounced when using R_b as the core radius measure. The physical origin of this dependence on the binary pericentre is not yet clear, and we do not identify a specific mechanism that could account for the additional predictive power of $r_{\text{peri},b}$.

Equations (d4)–(d6) repeat the same analysis using the core-Sérsic R_b as an alternative measure of the core radius. The resulting relations are consistent with those obtained for R_γ , but systematically exhibit lower coefficients of determination. This may partly reflect the additional parameter degeneracies inherent to the core-Sérsic fit, which introduce greater uncertainty in the determination of R_b , or that other galaxy properties may have a stronger influence on R_b than on R_γ , owing to the different definitions of these two core radius measures. Nevertheless, these relations recover two interesting scalings. First, we find $R_b \propto r_{\text{inf}}^{1.03 \pm 0.13}$, consistent with the approximately linear relation between core radius and influence radius reported by [Dosopoulou et al. \(2021\)](#) and [Thomas et al. \(2016\)](#). Second, we obtain $R_b \propto M_{\text{bin}}^{0.66 \pm 0.19}$, in agreement with the scaling between core radius and final SMBH mass found observationally by [Dullo & Graham \(2014\)](#).

Figure 11 shows the corresponding scaling relations for R_b , including the uncertainties on the fitted coefficients and the rms scatter around each relation. For the R_b – r_{inf} relation, we additionally show the measurements from [Thomas et al. \(2016\)](#), allowing a

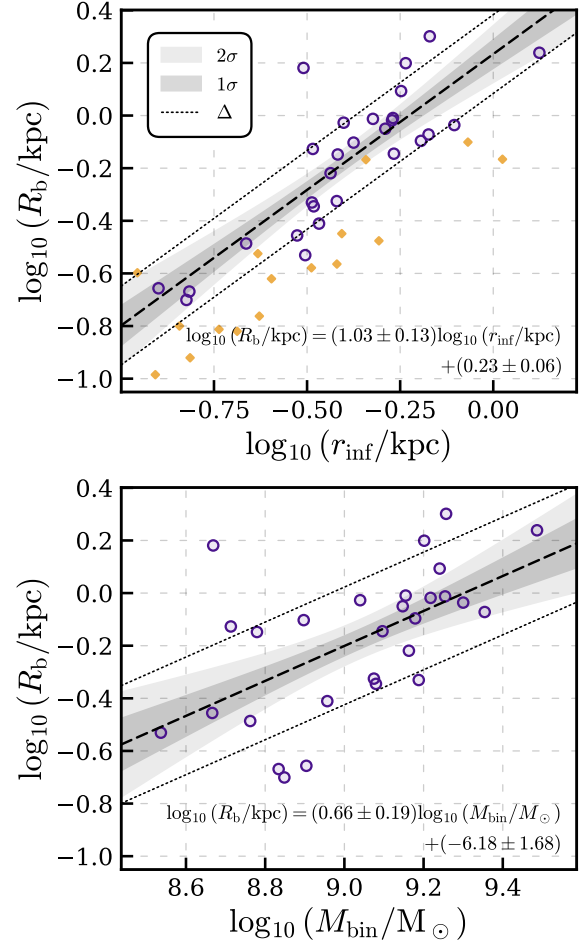


Figure 11. Scaling relations between the core size R_b and the binary influence radius, r_{inf} (top row) or total binary mass, M_{bin} (bottom row). The dashed lines show the best fitting relations, with the corresponding fit coefficients indicated in each panel. The shaded regions enclose the 1σ and 2σ confidence intervals associated with the fitted relations, while the dotted lines indicate the rms residual scatter. The yellow points show the measurements from [Thomas et al. \(2016\)](#) included for comparison in the R_b – r_{inf} relation.

direct comparison with the observational relation reported in that work. The observational measurements broadly overlap with our simulated remnants, with a very similar slope but a slight offset, such that our remnants tend to have somewhat larger core radii than the observed galaxies. This offset is also found by [Dosopoulou et al. \(2021\)](#) and increases as the initial inner density slope of the progenitors becomes shallower. Since our progenitors generally have relatively shallow central profiles, this may contribute to the larger core radii of our remnants relative to the observed galaxies.

Relations for the inner slope. The parameter carrying the most information on the inner density slope of the remnant at $a = a_h/3$ is the initial eccentricity of the galaxy encounter, e_0 . Relation (e1) in Table 3 provides the corresponding single-predictor scaling, with $R^2 = 0.43$. More eccentric encounters tend to produce shallower, more flattened central density slopes. This dependence is at least partly driven by the direct impact of the encounter orbit on the central structure of the remnant, independently of the presence of SMBHs. The merger induces central heating, which lowers the central density and consequently flattens the inner density profile. Indeed, a similar

relation is recovered in the simulations without black holes, given by relation (e2). Relation (e3) extends the previous scaling by including the black hole mass of the first progenitor, $M_{\text{BH,FP}}$, as a second predictor, increasing the coefficient of determination to $R^2 = 0.66$. Including information about the secondary black hole as well, by replacing $M_{\text{BH,FP}}$ with the binary mass M_{bin} and the black hole mass ratio q_{BH} , yields a three-parameter relation with $R^2 = 0.71$. Although the partial F -test gives a p -value of 0.055⁸, slightly above our adopted significance threshold of $p = 0.05$, the proximity to this threshold and the indication that the masses of both black holes may contribute to predicting the inner density slope motivate us to retain this relation for completeness. Taken together, these results suggest that the inner density slope is set primarily by the orbital configuration of the merger, with the SMBH properties providing a secondary contribution.

3.2.2 Mass deficit and ejected mass

Mass deficit. We first assess whether the conventional mass deficit, M_{def} , provides a reliable measure of the mass ejected by the SMBH binary. The left-hand panel of Fig. 12 shows the correlation between M_{def} and the maximum ejected mass, $M_{\text{ej}}^{\text{max}}$, both normalised by the binary mass. The correlation is relatively weak, with the corresponding relation in equation (f1) yielding $R^2 = 0.38$ and a substantial scatter. This is surprisingly low given that M_{def} is commonly used to characterise the depletion of the central stellar component induced by SMBH binary interactions.

The weak correspondence arises because the mass deficit does not directly trace the mass removed by the binary. As illustrated by the profiles in Fig. 9, the mass deficit inferred from the extrapolation of the outer Sérsic profile can substantially exceed the actual difference between the final density profiles with and without black holes. The six remnants shown in the figure span $M_{\text{def}}/M_{\text{bin}} \simeq 2.19\text{--}27.81$, whereas the maximum ejected mass remains in the much narrower range $M_{\text{ej}}^{\text{max}}/M_{\text{bin}} \simeq 0.45\text{--}1.74$. Interestingly, similarly large values of the mass deficit are also found observationally: Rusli et al. (2013) report $M_{\text{def}} \simeq 0.2\text{--}39 M_{\text{BH}}$ for their sample of core galaxies. The large discrepancy between M_{def} and $M_{\text{ej}}^{\text{max}}$ in our simulations indicates, however, that a substantial fraction of the inferred mass deficit is not directly associated with stellar ejection by the binary, but rather reflects the difficulty of accurately estimating what the galaxy would have looked like in the absence of binary scouring.

Ejected mass. We next investigate scaling relations for the mass ejected by the SMBH binary. The strongest predictor is the binary mass, M_{bin} , with the corresponding relation (g1) yielding $R^2 = 0.66$. We find $M_{\text{ej}}^{\text{max}} \propto M_{\text{bin}}^{1.13 \pm 0.16}$ which is consistent with the approximately linear dependence expected from the theoretical scaling

$$M_{\text{ej}}(a_f) \propto J M_{\text{bin}} \ln \left(\frac{a_i}{a_f} \right), \quad (19)$$

where J is the dimensionless hardening coefficient. For fixed values of J and a_i/a_f , this relation predicts an approximately linear scaling of the ejected mass with the binary mass.

Interestingly, the initial eccentricity of the galaxy encounter, e_0 ,

also provides non-negligible predictive power, with the corresponding relation (g2) yielding $R^2 = 0.38$. Higher encounter eccentricities are associated with larger values of $M_{\text{ej}}^{\text{max}}$. This dependence may arise from the impact of the merger orbit on the orbital structure of the remnant. As shown above, e_0 strongly influences the remnant shape, rotational support, velocity anisotropy, and inner density slope, which in turn modify the population of low angular momentum orbits that can reach the central region and interact with the SMBH binary. To explore this possibility, we compute at t_b the stellar mass M_{lc} contained in the loss cone, defined as the total stellar mass with angular momentum below the loss-cone angular momentum L_{lc} . We define L_{lc} as the maximum specific angular momentum for which a stellar orbit can reach a distance smaller than the binary semi-major axis a ,

$$L_{\text{lc}} = a \sqrt{2[E - \Phi(a)]}, \quad (20)$$

where E is the specific orbital energy and Φ is the gravitational potential.⁹

Figure 13 shows M_{lc} as a function of the initial merger eccentricity¹⁰. We find a mild trend for more eccentric encounters to produce remnants with a larger mass of stars able to reach the central region. These are also the encounters that produce more prolate, less rotationally supported, and more radially anisotropic remnants. This provides a link between the initial orbital eccentricity and the enhanced ejected mass: more eccentric mergers modify the remnant orbital structure in a way that leaves a larger reservoir of low angular momentum stars available for interaction with the SMBH binary.

We also find a significant correlation of $M_{\text{ej}}^{\text{max}}$ with the stellar velocity dispersion measured within the stellar half-mass radius at t_b , σ_{hm} , before the onset of SMBH binary scouring. The corresponding relation (g3) gives $R^2 = 0.57$. The correlation with σ_{hm} is not unexpected, since this quantity encodes information on several properties relevant to SMBH binary evolution. It enters implicitly into our definition of the final binary separation, $a_f = a_b/3$, since a_b depends on σ (Merritt 2006). In addition, σ_{hm} correlates with the central density and enclosed stellar mass of the remnant, as discussed in François et al. (2026). It is also correlated with the binary mass in our sample, with $R^2 = 0.50$, as more massive SMBHs tend to reside in systems with larger velocity dispersions. Finally, σ_{hm} increases with the initial encounter eccentricity, with $R^2 = 0.23$. The predictive power of σ_{hm} therefore likely reflects its correlation with several underlying properties that influence the amount of stellar mass interacting with and eventually ejected by the SMBH binary, rather than a single direct causal dependence.

The best two-parameter relation combines the binary mass with the initial eccentricity of the galaxy encounter, yielding $R^2 = 0.79$ (relation g4). The best three-parameter relation further includes the density of stars originating from the first progenitor, measured within the binary influence radius at binary formation, $\rho_{\text{FP},t_b}(r_{\text{inf}})$ (relation g5). This relation provides a further improvement in predictive power, with $R^2 = 0.88$. Interestingly, a closely related relation is obtained when replacing this quantity with the total remnant density within r_{inf} (relation g6), but the progenitor specific density provides the stronger predictor.

⁹ The resulting trends are qualitatively unchanged for loss-cone radii of $2a$ and $3a$.

¹⁰ The values of M_{lc} should not be interpreted as the stellar mass that will ultimately be ejected by the binary, since they are measured at t_b , when dynamical friction can still contribute to further binary hardening independently of stellar scouring. The quantity shown here is instead intended as an indication of the amount of stellar mass able to reach the central region at the onset of the scouring phase.

⁸ Under the null hypothesis that the additional predictor does not provide a genuine improvement over the two-parameter relation, a p -value of 0.055 indicates that an improvement at least as large as that observed would occur in approximately 5.5% of equivalent samples due to statistical fluctuations alone.

Table 3. Scaling relations for the core properties of the merger remnants. The table is divided into four groups corresponding to the core radius, inner slope, mass deficit, and ejected mass. For each relation we report: the number of predictors n_{var} ; the coefficient of determination R^2 ; the adjusted coefficient of determination R^2_{adj} , which penalises model complexity and allows fair comparison across relations with different numbers of predictors; the intrinsic scatter Δ ; the maximum variance inflation factor (VIF), which quantifies multicollinearity among predictors; and the root mean square error of a leave-one-out cross validation (LOOCV), which estimates out of sample predictive performance.

Relation		n_{var}	R^2	R^2_{adj}	Δ	VIF	LOOCV
Core radius							
d1	$\log_{10}(R_{\gamma}/\text{kpc}) = +0.85 \log_{10}(r_{\text{inf}}/\text{kpc}) + 0.01$	1	0.93	0.93	0.05	$\times$	0.27
d2	$\log_{10}(R_{\gamma}/\text{kpc}) = +0.55 \log_{10}(M_{\text{bin}}/M_{\odot}) - 5.30$	1	0.44	0.42	0.15	$\times$	0.79
d3	$\log_{10}(R_{\gamma}/\text{kpc}) = +0.85 \log_{10}(r_{\text{inf}}/\text{kpc}) + 0.06 \log_{10}(r_{\text{peri,b}}/\text{kpc}) + 0.14$	2	0.95	0.95	0.04	1.0	0.24
d4	$\log_{10}(R_{\text{b}}/\text{kpc}) = +1.03 \log_{10}(r_{\text{inf}}/\text{kpc}) + 0.23$	1	0.70	0.69	0.15	$\times$	0.57
d5	$\log_{10}(R_{\text{b}}/\text{kpc}) = +0.66 \log_{10}(M_{\text{bin}}/M_{\odot}) - 6.18$	1	0.33	0.30	0.22	$\times$	0.87
d6	$\log_{10}(R_{\text{b}}/\text{kpc}) = +1.03 \log_{10}(r_{\text{inf}}/\text{kpc}) + 0.17 \log_{10}(r_{\text{peri,b}}/\text{kpc}) + 0.62$	2	0.78	0.76	0.13	1.0	0.51
Inner slope							
e1	$\gamma = -0.11 e_0 + 0.21$	1	0.43	0.40	0.02	$\times$	0.86
e2	$\gamma_{\text{noBH}} = -0.10 e_0 + 0.18$	1	0.32	0.29	0.02	$\times$	0.86
e3	$\gamma = -0.15 e_0 + 0.05 \log_{10}(M_{\text{BH,FP}}/M_{\odot}) - 0.16$	2	0.66	0.63	0.01	1.1	0.74
e4	$\gamma = -0.15 e_0 + 0.04 \log_{10}(M_{\text{bin}}/M_{\odot}) + 0.01 q_{\text{BH}} - 0.09$	3	0.71	0.67	0.01	1.2	0.70
Mass deficit							
f1	$\log_{10}(M_{\text{def}}/M_{\odot}) = +0.83 \log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) + 2.32$	1	0.38	0.36	0.35	$\times$	0.84
Ejected mass							
g1	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +1.13 \log_{10}(M_{\text{bin}}/M_{\odot}) - 1.30$	1	0.66	0.65	0.19	$\times$	0.64
g2	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +1.52 e_0 + 7.58$	1	0.38	0.36	0.26	$\times$	0.84
g3	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +3.81 \log_{10}(\sigma_{\text{hm}}/[\text{km s}^{-1}]) + 11.03$	1	0.57	0.55	0.22	$\times$	0.72
g4	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +0.95 \log_{10}(M_{\text{bin}}/M_{\odot}) + 0.94 e_0 - 0.49$	2	0.79	0.78	0.15	1.1	0.51
g5	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +1.11 \log_{10}(M_{\text{bin}}/M_{\odot}) + 0.73 e_0$ $+0.15 \log_{10}(\rho_{\text{FP,tb}}(r_{\text{inf}})/[M_{\odot} \text{ kpc}^{-3}]) - 3.16$	3	0.88	0.86	0.11	1.3	0.40
g6	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +1.13 \log_{10}(M_{\text{bin}}/M_{\odot}) + 0.75 e_0$ $+0.16 \log_{10}(\rho_{\text{rem,tb}}(r_{\text{inf}})/[M_{\odot} \text{ kpc}^{-3}]) - 3.49$	3	0.84	0.82	0.13	1.5	0.40
g7	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +1.09 \log_{10}(M_{\text{bin}}/M_{\odot}) + 1.01 e_0$ $-0.13 \log_{10}(M_{\text{DM,SP}}/M_{\odot}) - 0.28$	3	0.87	0.85	0.12	1.3	0.43
g8	$\log_{10}(M_{\text{ej}}^{\text{max}}/M_{\odot}) = +0.19 \log_{10}(M_{\text{def}}/M_{\odot}) + 2.72 \log_{10}(\sigma_{\text{hm}}/[\text{km s}^{-1}])$ $+0.01 \log_{10}(R_{\text{e}}/\text{kpc}) - 1.44 \gamma + 8.61$	4	0.75	0.70	0.17	1.7	0.68

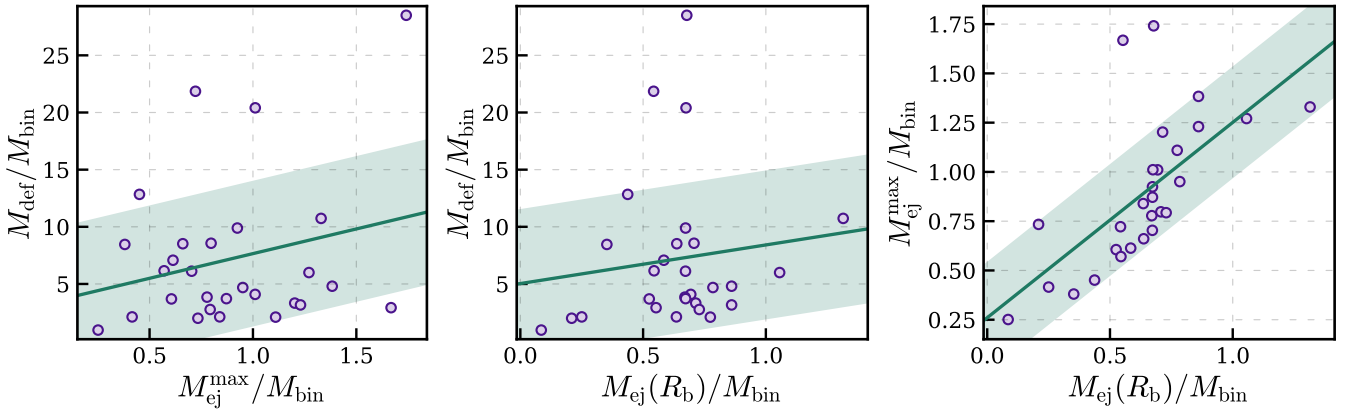


Figure 12. Comparison between the classical mass deficit, M_{def} , and the mass ejected from the center of the remnants. The left panel shows M_{def} as a function of the maximum ejected mass, $M_{\text{ej}}^{\text{max}}$, while the middle panel shows M_{def} as a function of the mass ejected within the core-Sérsic break radius, $M_{\text{ej}}(R_{\text{b}})$. The right panel shows the relation between $M_{\text{ej}}^{\text{max}}$ and $M_{\text{ej}}(R_{\text{b}})$. All quantities are normalised by the binary mass, M_{bin} . Solid lines show the best fitting linear relations, with the shaded regions indicating the scatter of the corresponding relations. The weak correlation between M_{def} and $M_{\text{ej}}^{\text{max}}$ demonstrates that the classical mass deficit provides a poor estimate of the actual mass ejected by the binary.

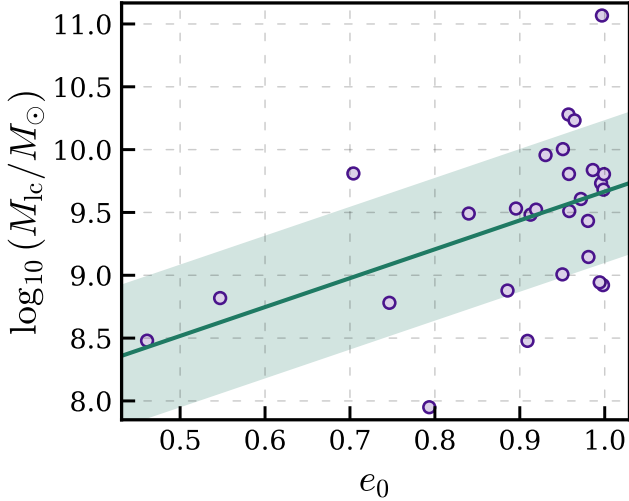


Figure 13. Total stellar mass on centrophilic orbits, M_{Lc} , defined as the mass of stars with angular momentum $L < L_{\text{Lc}}$, as a function of the initial orbital eccentricity e_0 of the galaxy merger. The solid line shows the best-fitting linear relation, with the shaded region indicating the scatter of the relation. More eccentric mergers tend to produce remnants with a larger population of centrophilic orbits.

To assess whether this difference reflects a genuine improvement rather than statistical fluctuations, we first remove the dependence on M_{bin} and e_0 from each quantity (M_{ej} , $\rho_{\text{FP,tb}}(r_{\text{inf}})$ and $\rho_{\text{rem,tb}}(r_{\text{inf}})$) by computing the residuals of the corresponding two-parameter relations, namely $M_{\text{ej}} - (M_{\text{bin}}, e_0)$, $\rho_{\text{FP,tb}}(r_{\text{inf}}) - (M_{\text{bin}}, e_0)$, and $\rho_{\text{rem,tb}}(r_{\text{inf}}) - (M_{\text{bin}}, e_0)$. We then compute the Pearson correlations between the residuals of the ejected mass and those of the two density measures, and between the densities themselves too. The residuals of M_{ej} correlate more strongly with those of $\rho_{\text{FP,tb}}(r_{\text{inf}})$, with $r = 0.65$ ($p = 2.1 \times 10^{-4}$), than with those of the total remnant density, for which $r = 0.48$ ($p = 9.9 \times 10^{-3}$). To assess the statistical significance of the difference between the two dependent partial correlations we use Williams (1959) t -test¹¹ as implemented in the cocor package (Diedenhofen & Musch 2015), which is recommended for small samples. This yields $t(25) = 2.96$, $p = 0.007$, confirming that the partial correlation with $\rho_{\text{FP,tb}}(r_{\text{inf}})$ is significantly stronger than that with $\rho_{\text{rem,tb}}(r_{\text{inf}})$ at the 1% level. This result is robust across all nine tests implemented in cocor, which yield p -values ranging from 0.007 to 0.015. The 95% confidence interval for the difference in correlations, computed following Zou (2007), is $[0.038, 0.375]$, excluding zero¹². This indicates that the superiority of $\rho_{\text{FP,tb}}(r_{\text{inf}})$ is not an artefact of the limited sample size but reflects a genuine difference in predictive power. The stronger predictive power of the progenitor specific density is physically suggestive: it indicates that the central stellar population originating from the first progenitor may contribute preferentially to the stellar mass ejected by the binary. The density of these stars at binary formation therefore appears to provide more

direct information on the reservoir of stars subsequently interacting with the binary than the total central density of the remnant.

A practical limitation is that $\rho_{\text{FP,tb}}(r_{\text{inf}})$ is difficult to determine in cosmological simulations, where individual stellar populations are generally not tracked at this level of detail, and is not directly accessible observationally. We therefore seek a more readily available proxy. The dark matter mass of the second progenitor is particularly convenient, as it can be readily extracted from cosmological simulations. We find that it correlates with both $\rho_{\text{FP,tb}}(r_{\text{inf}})$ and $\rho_{\text{rem,tb}}(r_{\text{inf}})$, with more massive dark matter haloes in the second progenitor associated with lower central densities in the remnant. This may arise because a more massive dark matter halo in the second progenitor can induce stronger merger-driven heating of the central region of the first progenitor. This effect is enhanced by the fact that, although our systems are all major mergers in terms of stellar mass, the dark matter mass ratios span a broader range, with the dark matter component dominating the total progenitor mass (François et al. 2026). The dark matter mass of the second progenitor may therefore provide a proxy for the central density after the merger. Replacing ρ with $M_{\text{DM,SP}}$ yields a relation with $R^2 = 0.87$ (equation g7). This relation performs better than the corresponding relation using the total remnant density, likely because $M_{\text{DM,SP}}$ correlates more strongly with $\rho_{\text{FP,tb}}(r_{\text{inf}})$ than with $\rho_{\text{rem,tb}}(r_{\text{inf}})$. However, this difference is not statistically significant.¹³

Having identified the main physical parameters governing the ejected mass, we now seek a practical relation that can be used observationally. In particular, since the conventional mass deficit, M_{def} , is commonly used as an observable proxy for the mass ejected by the SMBH binary, we investigate whether its predictive power can be improved by incorporating additional quantities that are readily accessible from observations. Relation (g8) provides such a scaling, yielding a more accurate estimate of the ejected mass than M_{def} alone. Starting from M_{def} , which yields $R^2 = 0.38$, the addition of three observable quantities increases the coefficient of determination to $R^2 = 0.75$. These additional predictors are the inner density slope, γ , and effective radius, R_e , obtained from the core-Sérsic fit, together with the one-dimensional stellar velocity dispersion measured within the stellar half-mass radius. This relation therefore provides a practical alternative to using M_{def} alone to estimate the mass ejected by the SMBH binary.

Figure 14 shows the probability density distribution of the maximum ejected mass normalised by the binary mass, $M_{\text{ej}}^{\text{max}}/M_{\text{bin}}$. The distribution peaks at $M_{\text{ej}}^{\text{max}}/M_{\text{bin}} = 0.77$, with a median value of 0.82. The peak is broadly consistent with the expectation of ~ 0.50 for a single merger in the prescription of Merritt (2006). However, our distribution is broad, spanning $M_{\text{ej}}^{\text{max}}/M_{\text{bin}} \approx 0.25$ – 1.74 , indicating substantial variation in the amount of stellar mass ejected relative to the binary mass across our merger remnants.

4 CONCLUSIONS

4.1 Summary

We have investigated the properties of remnants produced by dry major mergers of massive elliptical galaxies with SMBH at low redshift, a population expected to contribute significantly to the nanohertz gravitational wave signal targeted by Pulsar Timing Arrays. Our aim is to understand how such mergers shape the global properties of the

¹¹ The t -statistic measures how large the observed difference between the two correlations is relative to the uncertainty expected under H_0 .

¹² The 95% confidence interval provides an estimate of the range within which the true difference between the two correlations lies. Since zero is not included in the interval, the two correlations are statistically significantly different at the 5% level.

¹³ A Williams t -test for dependent correlations gives $t(25) = -1.52$ and $p = 0.14$, while the 95% confidence interval for the difference, computed following Zou (2007), is $[-0.28, 0.05]$ and includes zero.

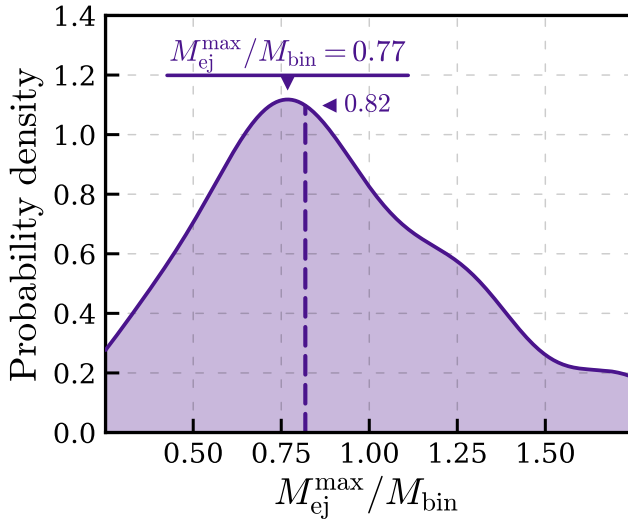


Figure 14. Probability density distribution of the maximum stellar mass ejected by the SMBH binary, normalised by the binary mass, $M_{\text{ej}}^{\text{max}}/M_{\text{bin}}$. The vertical dashed line indicates the median value of $M_{\text{ej}}^{\text{max}}/M_{\text{bin}} = 0.82$, while the marker denotes the peak of the distribution at $M_{\text{ej}}^{\text{max}}/M_{\text{bin}} = 0.77$.

resulting galaxies, including their intrinsic shape and stellar kinematics, as well as their central cores, and to identify the parameters that primarily govern these properties. To this end, we combine state of the art high-resolution N -body simulations with progenitor galaxy structures and merger orbits directly extracted from cosmological simulations, providing initial conditions that retain the diversity and complexity of the galaxy population and merger configurations expected in a cosmological context.

We first investigate the relations between the different structural and kinematic properties of the remnants, and between these properties and the progenitor and merger parameters that set their initial conditions. We then turn to the formation of stellar cores and the associated stellar mass ejection in the same cosmological merger framework, where the initial conditions span a range of galaxy structures and orbital configurations beyond those explored in idealised merger models. For each aspect, we derive empirical scaling relations that provide predictions for the shape, kinematics, core properties, and stellar scouring of elliptical galaxy merger remnants.

The **main results** can be summarised as follows:

■ **Shape and kinematics of ETGs.** The intrinsic shape, rotational support, and velocity anisotropy of the merger remnants are strongly determined by the orbital properties of the encounter that formed them. Among the parameters considered, the initial eccentricity e_0 is the dominant predictor, while the strongest two-parameter relations are obtained when the initial semi-major axis a_0 is included as a second predictor. These dependencies also give rise to strong correlations between the different structural and kinematic properties. More eccentric encounters produce more prolate remnants, with weaker rotational support and stronger radial velocity anisotropy, while the opposite trend is found for less eccentric mergers. Given the strongly eccentric orbits typical of cosmological galaxy mergers, a representative remnant is therefore expected to be highly prolate ($T \sim 0.95$), weakly rotating ($v_\phi/\sigma_\phi \sim 0.03$), and radially anisotropic ($\beta_{\text{max}} \sim 0.5$). We provide practical scaling relations linking the remnant shape, rotation, anisotropy and initial merger orbit. They can

therefore be used to set these properties in more realistic initial models by specifying the merger eccentricity, with $e_0 \approx 0.94$ providing a cosmologically representative default value; the resulting T , v_ϕ/σ_ϕ , and β can then be used, for example, to construct tailored initial conditions with Schwarzschild modelling. The scaling relations further indicate that encounters with $e_0 \gtrsim 0.65$ produce prolate remnants with $T > 0.5$.

■ **Comparison with ILLUSTRISTNG galaxies.** The merger remnants produced by our high-resolution N -body simulations are broadly consistent with the ILLUSTRISTNG early-type galaxy population in terms of their intrinsic shape and rotational support. Their global velocity anisotropy is, however, systematically more radially biased. The presence of gas in ILLUSTRISTNG accounts for part of this offset, but does not fully explain it. The remaining difference likely reflects the more complex evolutionary and environmental processes present in the cosmological simulations, whereas our collisionless simulations isolate the dynamical imprint of the merger itself.

■ **Cores.** The stellar surface density profiles of remnants formed with and without an SMBH binary are broadly similar, with no significant systematic difference in their inner slopes. This similarity arises because the remnants largely inherit the relatively shallow central density profiles of their progenitors, whose three-dimensional inner slopes are generally below $\gamma_{\text{pr},3\text{D}} = 1$, with the majority lying in the range $\gamma_{\text{pr},3\text{D}} \approx 0.4$ – 0.7 . In projection, these correspond to inner slopes below $\gamma_{\text{pr},2\text{D}} \sim 0.3$. The shallow density profile inherited from the progenitor is further flattened by merger-driven central heating. Subsequent SMBH binary scouring primarily affects the slope of remnants with projected inner slopes $\gamma \gtrsim 0.15$, reducing them to $\gamma \approx 0.05$ – 0.15 . At still shallower slopes, continued binary scouring mainly extends the depleted region to larger radii, leading to a progressive increase in the core radius. The resulting core radii span $R_b \approx 0.2$ – 2 kpc, with inner slopes of $\gamma \approx 0.05$ – 0.15 . We recover the previously reported relation $R_b \approx r_{\text{inf}}$, with our best fit relation giving $R_b \propto r_{\text{inf}}^{1.03 \pm 0.13}$. The prediction of R_b is further improved by including the SMBH binary pericentre prior to the scouring phase. We also recover the observationally motivated correlation between core radius and binary mass, with $R_b \propto M_{\text{bin}}^{0.66 \pm 0.19}$, although with substantially weaker predictive power than the relation based on r_{inf} . The inner slope is most strongly correlated with the initial merger eccentricity: more eccentric encounters produce flatter cores. This dependence is, at least in part, driven by the direct effect of merger induced heating on the central density. Once the inner profile has already been substantially flattened by the shallow progenitor profiles and the merger itself, further flattening by SMBH binary hardening is relatively modest, since it becomes progressively more difficult to reduce an already shallow inner slope. The SMBH binary therefore provides a secondary contribution to the evolution of γ , which is reflected in the scaling relations.

■ **Effect of SMBHs.** The presence of the SMBH binary has little impact on the global properties of the merger remnants, with no significant differences in their shape, rotational support, or velocity anisotropy on galactic scales. Within the central region, however, the presence of the binary systematically lowers the triaxiality parameter. The rotational support remains largely unaffected by the binary, while the velocity anisotropy is significantly modified. The SMBH binary preferentially ejects stars on radial orbits through three-body slingshot interactions, producing a tangentially biased central region, while the corresponding remnants without SMBHs remain mildly radially biased. Although cores are also present in the absence of SMBHs, reflecting the relatively shallow initial density profiles, merger-induced central heating, and the absence of gas dissipation,

the presence of the SMBH binary leads to systematically larger core radii and shallower inner density slopes.

■ **Mass deficit and ejected mass.** Our mass deficits are consistent with the range of values typically inferred observationally. However, the conventional mass deficit used in the literature as a proxy for SMBH binary scouring correlates only weakly with the stellar mass actually ejected by the binary ($R^2 = 0.38$), with substantial scatter. We provide an observationally motivated scaling relation that improves the prediction of the mass deficit by combining the one-dimensional velocity dispersion with the core-Sérsic parameters R_e and γ , increasing the predictive power from $R^2 = 0.38$ to $R^2 = 0.75$. The actual ejected mass depends primarily on the binary mass and initial merger eccentricity, with the latter affecting the size of the centrophilic orbital population available for ejection. We also find a correlation with the remnant velocity dispersion. We derive two- and three-parameter scaling relations linking the ejected mass to the progenitor and orbital properties. The best two-parameter relation uses M_{bin} and e_0 , while the best three-parameter relation additionally includes the dark matter mass of the second progenitor. We find that this quantity provides a good proxy for the central density of the remnant, which is closely connected to the amount of stellar mass available for ejection. A larger dark matter mass in the second progenitor is associated with a lower central density in the remnant. We further show that stars originating from the first progenitor are preferentially ejected by the SMBH binary. Finally, the ratio $M_{\text{ej}}/M_{\text{bin}}$ spans 0.25–1.74, with a peak at $M_{\text{ej}}/M_{\text{bin}} \approx 0.77$.

4.2 Comparison with previous work and discussion

Shape. Our results for the global shape of the merger remnants are broadly consistent with previous N -body studies of galaxy mergers, which have found that radial encounters preferentially produce prolate remnants (Villumsen 1982; Boylan-Kolchin et al. 2005; González-García & Van Albada 2005a,b; Boylan-Kolchin et al. 2006). They are also consistent with Bortolas et al. (2018), who found that the large scale remnant morphology is largely unaffected by the presence of an SMBH binary and that the triaxiality increases with the eccentricity of the merger orbit. Their remnants, however, have systematically lower values of T , with $T \lesssim 0.6$, whereas most of our remnants have $T > 0.6$. This difference is naturally related to the different merger initial conditions. First, the merger orbits in our cosmological sample are substantially more eccentric: while Bortolas et al. (2018) explored $e_0 = 0.5, 0.7$, and 0.9 , most of our encounters have $e_0 > 0.9$. Given our T – e_0 relation, $T > 0.6$ is expected for $e_0 \gtrsim 0.73$, consistent with the lower triaxialities found in their simulations. Second, their models consist of single-component stellar systems with equal mass progenitors, whereas our progenitors have nearly equal stellar mass but can have substantially different dark matter masses (see François et al. 2026), resulting in total merger mass ratios that can differ significantly from unity. Finally, our progenitors have central density slopes predominantly in the range $0.3 \lesssim \gamma_{\text{pr},3D} \lesssim 0.7$, whereas Bortolas et al. (2018) considered models with $\gamma_{\text{pr},3D} = 0.5, 1$, and 1.5 . This difference also acts in the direction expected from their results, as they find that less concentrated progenitors produce more triaxial remnants. Thus, both the higher eccentricities and lower central concentrations of our cosmological progenitors naturally favour the larger triaxialities found in our sample.

On smaller scales, our results show that the remnants become more nearly spherical towards the centre, with the presence of the SMBH binary systematically reducing the triaxiality parameter. This behaviour is consistent with the central shape results of Bortolas et al.

(2018) and with earlier studies showing that a central mechanism capable of efficiently scattering box orbits tends to suppress triaxiality and drive the stellar system towards a more spherical configuration (Gerhard & Binney 1985; Merritt & Fridman 1995; Merritt & Quinlan 1998). In our case, the SMBH binary provides such a central source of orbital scattering.

Bortolas et al. (2018) found no clear connection between the remnant morphology and the efficiency of SMBH binary hardening. We find a consistent result in our cosmological merger sample: using the hardening timescale Δt_h defined in François et al. (2026) and the hardening rate s , we find no significant correlation with the remnant shape, either globally or in the central regions.

At the cosmological level, our results are also consistent with Li et al. (2018), who studied massive galaxies in the ILLUSTRIS simulation and found that nearly all prolate systems had experienced at least one dry major merger, with most of these mergers occurring on nearly radial orbits.

Rotation. Our results are consistent with previous N -body and cosmological studies showing that the orbital configuration of a galaxy merger strongly influences the rotational support of the remnant (González-García & Van Albada 2005a,b; Moody et al. 2014; Bortolas et al. 2018; Li et al. 2018). The remnants of Bortolas et al. (2018) have substantially stronger rotational support, with typical values of $v/\sigma \sim 0.8$, which is consistent with our e_0 – v/σ relation given the lower eccentricities explored in their simulations. They also found that the presence of an SMBH binary has little effect on the rotational support at large radii, in agreement with our results. In the central regions, however, Bortolas et al. (2018) reported a difference in rotational support between simulations with and without SMBHs, whereas we find no significant change. This difference may depend on the time at which the rotation is measured. During the early stages of binary hardening, SMBH binary interactions preferentially eject prograde stars, temporarily increasing the fraction of retrograde orbits in the central region. As the binary continues to interact with the surrounding stars, their angular momenta are chaotically perturbed, allowing the retrograde orbits to be scattered into prograde ones. This process can gradually restore the balance between prograde and retrograde orbits, such that the net ordered rotation is not substantially altered at later times (see Fig. 4 of Iwasawa et al. 2011). The same angular-momentum exchange also contributes to the growth of the binary eccentricity found in our simulations (François et al. 2026). We therefore expect the impact of SMBH scouring on the net central rotation to depend on the time at which the remnant is analysed, with little long-term change expected for a system that initially has no net rotation.

Anisotropy. The tendency of galaxy mergers to produce radially biased stellar kinematics on galactic scales is consistent with previous numerical studies (González-García & Van Albada 2005a,b; Bournaud et al. 2007). Similarly, Bortolas et al. (2018) found that more radial encounters produce remnants with larger values of β . The tangential bias that develops in the central regions following SMBH binary evolution is also consistent with theoretical expectations and previous simulations (Bortolas et al. 2018; Rantala et al. 2018, 2024), as well as with observational measurements of massive early-type galaxies (Thomas et al. 2014).

Core properties. The dependence of the inner density slope on the initial merger eccentricity indicates that the merger orbit can influence the evolution of the central stellar density profile, consistent with a contribution from merger driven heating of the central regions.

Similar central heating during the merger has also been observed in previous numerical studies, using both direct N -body integrations (Bortolas et al. 2018) and tree-based simulations (Dosopoulou et al. 2021).

The core radii in our remnants span $R_b \approx 0.2$ – 2 kpc, with most systems lying between 0.2 and 1 kpc, in good agreement with the ranges reported observationally (Ravindranath et al. 2002; Rusli et al. 2013; Dullo et al. 2017; Dullo 2019). Our non-parametric core radii, $R_\gamma \approx 0.17$ – 1.13 kpc, are likewise consistent with the measurements of Lauer et al. (2007), who find values extending to ~ 0.8 kpc and a maximum of ~ 1.5 kpc¹⁴. The sizes of our cores are therefore broadly consistent with those observed in the local population of core ETGs. They are also consistent with what can be produced by SMBH binary scouring alone, without requiring additional mechanisms. Khonji et al. (2024) find that SMBH binary interactions can produce cores extending to ~ 1.3 kpc, whereas cores larger than ~ 2 kpc are more naturally associated with gravitational wave recoil. Most of our remnants have $R_b < 1.3$ kpc, while only four systems have core radii between 1.4 and 2 kpc and none exceed 2 kpc.

Scaling relations. For the scaling relations between the core and SMBH properties, we recover the relation $R_b \approx r_{\text{inf}}$ reported observationally by Thomas et al. (2016) and also found in the numerical study of Dosopoulou et al. (2021). Our best-fitting relation, $R_b \propto r_{\text{inf}}^{1.03 \pm 0.13}$ (relation d4), is remarkably close to the $R_b \propto r_{\text{inf}}^{1.05}$ scaling obtained by Dosopoulou et al. (2021). While the slope is also consistent with the observational relation of Thomas et al. (2016), we find a systematic offset in its normalisation. A similar offset is also found in the simulations of Dosopoulou et al. (2021) for models with central progenitor slopes in the range $\gamma_{\text{pr},3D} = 0.5$ – 1.5 , although their results suggest that the offset decreases slightly towards larger values of $\gamma_{\text{pr},3D}$.

The relation found by Rantala et al. (2018) is shallower and depends on the initial stellar density slope of the progenitors, with $R_b \propto r_{\text{inf}}^{0.39}$ for $\gamma_{\text{pr},3D} = 1$ and $R_b \propto r_{\text{inf}}^{0.74}$ for $\gamma_{\text{pr},3D} = 1.5$. They interpreted this dependence as support for a scenario in which the progenitors of massive core ellipticals are steep, cuspy galaxies with $\gamma_{\text{pr},3D} \gtrsim 3/2$. Our results show that the observed R_b – r_{inf} relation can nevertheless be reproduced with a population that includes shallower progenitor profiles. Similarly, Dosopoulou et al. (2021) found no strong variation of the R_b – r_{inf} relation with the initial density profile slope, in contrast to the dependence reported by Rantala et al. (2018).

The different behaviour may partly reflect the different ranges of initial conditions explored in the two studies. Rantala et al. (2018) kept the galaxy mass fixed while varying the SMBH mass over an order of magnitude and the progenitor density slope, whereas Dosopoulou et al. (2021) adopted a fixed SMBH mass fraction and varied the progenitor density slope and binary mass ratio. In our cosmological sample, both the progenitor density structure and the SMBH-to-galaxy mass ratio vary simultaneously over the range naturally produced by the cosmological population. We therefore do not need to select a particular initial density slope or SMBH-to-galaxy mass ratio to recover the R_b – r_{inf} relation, suggesting that the near-linear scaling is not restricted to a specific class of idealised progenitors.

¹⁴ The R_γ values reported by Lauer et al. (2007) are derived from Nuker-law fits. The resulting Nuker parameters, and hence R_γ , can depend on the radial extent adopted for the fit, so this comparison should be interpreted with some caution (Graham et al. 2003).

The R_b – M_{BH} relation found by Rantala et al. (2018) is also shallower than ours, with $R_b \propto M_{\text{BH}}^{0.23}$ for $\gamma_{\text{pr},3D} = 1$ and $R_b \propto M_{\text{BH}}^{0.52}$ for $\gamma_{\text{pr},3D} = 1.5$, compared with $R_b \propto M_{\text{bin}}^{0.66 \pm 0.19}$ in our simulations, despite our progenitors having substantially shallower inner density profiles. Using direct measurements of M_{BH} and estimates based on σ and L , respectively, Dullo & Graham (2014) find slopes of 0.83 ± 0.21 , 0.87 ± 0.18 , and 0.83 ± 0.21 , respectively, for the R_b – M_{BH} relation. Despite their somewhat steeper central values, all three measurements are consistent with our slope of 0.66 ± 0.19 within the combined uncertainties. More recently, Dullo (2019) found $R_b \propto M_{\text{BH}}^{0.83 \pm 0.10}$, also consistent with our relation.

Mass deficit. The mass deficits measured in our remnants span $M_{\text{def}} \approx 0.97$ – $28.5 M_{\text{bin}}$, covering the broad range reported observationally, from $M_{\text{def}} \sim 0.2$ – $39 M_{\text{BH}}$ in Rusli et al. (2013) to values of up to $\sim 20 M_{\text{BH}}$ in Dullo & Graham (2014). These comparisons should nevertheless be treated with caution, since the inferred mass deficit depends sensitively on the adopted parametrisation of the stellar profile. For example, Dullo & Graham (2013) argued for substantially lower values, $M_{\text{def}} \sim 0.5$ – $2 M_{\text{BH}}$, highlighting the systematic differences between different approaches. Dosopoulou et al. (2021) likewise emphasised that mass deficits derived from an assumed initial non-cored profile should be regarded primarily as an order of magnitude estimate of the ejected mass, since there is no reason to expect a universal progenitor density profile. More fundamentally, Hopkins & Hernquist (2010) showed that an extended outer envelope generated by previous mergers can alter the Sérsic fit and produce an apparent central mass deficit even when the central density profile has undergone little change. Similar difficulties are observed by Rantala et al. (2024) when the outer parts of merger remnants approach a power law profile, while Rusli et al. (2013) identified analogous issues observationally, motivating more careful or multi-component modelling. This illustrates that M_{def} should not necessarily be interpreted as a direct measure of the stellar mass ejected by an SMBH binary, and is consistent with our finding of only a weak correlation between M_{def} and M_{ej} .

Ejected mass. The ejected stellar mass spans $M_{\text{ej}} = 0.25$ – $1.74 M_{\text{bin}}$, with a peak at $0.77 M_{\text{bin}}$ and a median of $0.82 M_{\text{bin}}$. This range is broadly consistent with the characteristic value reported by Merritt (2006), although their definition of the ejected mass differs from ours, and with the 0.7 – $1.6 M_{\text{bin}}$ range found by Dosopoulou et al. (2021). The latter, however, estimated the ejected mass using two different approaches: either by comparing the density profiles before and after scouring, or by comparing the final remnant with the primary progenitor. We tested similar prescriptions, including a temporal comparison of the density profiles before and after SMBH binary evolution, but found that the inferred mass depends strongly on the radius at which the enclosed mass difference is evaluated. This method is also affected by residual non-equilibrium evolution after the galaxy merger, which can drive mass redistribution unrelated to the SMBH binary. We additionally considered an energy kick based estimate, but found it difficult to distinguish the small energy changes associated with the post-merger relaxation from those produced by interactions with the binary. These tests indicate that accurately isolating the mass ejected by the SMBH binary is challenging unless the particle trajectories can be followed with sufficient temporal resolution to identify the stars that interact with the binary and receive an energy kick, or a control simulation identical in every respect except for the presence of SMBHs is available. The latter approach, adopted here, provides a direct way to isolate the contribution of the SMBH binary to the stellar mass redistribution.

4.3 Caveats

Our progenitor density profiles are obtained by fitting the stellar and dark matter distributions of the ILLUSTRISTNG galaxies. An advantage of this approach is that the progenitor profiles retain the imprint of the complex environmental processes present in the cosmological simulation, including feedback processes that are absent from idealised isolated galaxy models. However, these profiles are also subject to the finite spatial resolution of ILLUSTRISTNG, and in particular to its relatively large gravitational softening length, which may affect their innermost regions. This limitation does not directly carry over to our resimulations, since the fitted profiles are used only to construct the initial conditions, which are then evolved at much higher resolution. The limited resolution of ILLUSTRISTNG may nevertheless result in progenitor profiles that are artificially shallow in their central regions, and therefore flatter than the cuspy profiles commonly adopted in idealised merger simulations. It is not, however, clear that imposing such idealised cusps would provide a more realistic description of the progenitors considered here. For the first generations of early-type galaxies, observational evidence for steep central cusps provides a motivation for such initial conditions. Our progenitors, in contrast, are already massive early-type galaxies at low redshift and may have experienced previous merger-driven SMBH scouring or gravitational wave recoil. Both processes can reduce the central stellar density and flatten the inner density profile, but their effects are not captured by the treatment of SMBHs in ILLUSTRISTNG. For example, repeated SMBH mergers can progressively flatten the central density profile (see Rantala et al. 2024). Previous mergers or recoil events could therefore have altered the central structure of our progenitors prior to the merger considered here. Similarly, environmental processes not fully captured in an idealised galaxy model may also modify the inner profile. Consequently, although the central slopes inferred from ILLUSTRISTNG may be biased towards lower values by numerical resolution, it is not obvious that the steeper cusps usually assumed in idealised models are closer to the actual progenitors of massive low redshift ellipticals. A choice of initial inner density profile is unavoidable in any resimulation of this kind. Rather than imposing an idealised cusp, we therefore use the fitted ILLUSTRISTNG profiles, thereby retaining their measured structural diversity. This choice also provides a more faithful representation of the global mass distribution. Indeed, fitting the ILLUSTRISTNG profiles with more restrictive models, such as Dehnen profiles, can lead to substantial mismatches in the total progenitor masses, with differences reaching $\sim 30\%$. Our more flexible density model instead recovers the progenitor masses to within $\sim 2\%$. We therefore consider it preferable to reproduce the cosmological progenitor profiles as accurately as possible rather than impose a priori inner cusps whose validity for massive low redshift ellipticals is uncertain.

We do not model gas dynamics in our simulations, and our results should therefore primarily be interpreted in the context of gas poor massive galaxies, which are expected to dominate the mergers contributing to the assembly of massive early-type galaxies at low redshift. Nevertheless, some massive systems may retain a non-negligible gas content, in which case dissipative processes and gas driven interactions could modify both the SMBH binary evolution and the resulting stellar structure. Such systems would require dedicated hydrodynamical modelling.

A further caveat concerns the treatment of the final stages of the binary evolution. Our N -body code does not include post-Newtonian corrections, and we therefore characterise the core properties at $a = a_h/3$. In reality, however, post-Newtonian effects will eventually become important at a separation a_{GW} that may not coincide

with $a_h/3$. If $a_{\text{GW}} < a_h/3$, our measurements are performed before the onset of the final relativistic evolution and may therefore underestimate the final core size. The evolution shown in Fig. B1 suggests that this effect should have little impact on the inner slope γ , which remains approximately constant during the hardening phase. We similarly find that the ejected mass M_{ej} is largely unaffected, as the bulk of the ejected mass is produced before a_h , in agreement with Sesana et al. (2006) (see their Fig. 5). For highly eccentric binaries, however, gravitational wave emission can become important at substantially larger separations, causing the binary to merge more rapidly and potentially reducing the amount of core scouring. This behaviour has been demonstrated by Rantala et al. (2024) (see their Fig. B1). We attempted to extrapolate the evolution of our core properties to a_{GW} , but for some mergers the available number of snapshots is insufficient to provide reliable estimates, resulting in large uncertainties. We therefore retain $a_h/3$ as our reference separation. We also identified two highly eccentric mergers for which the binary coalesces essentially immediately after binary formation (see François et al. 2026). These two systems were therefore excluded from the initial sample of 30 mergers, as their extremely rapid evolution is not expected to produce significant SMBH binary scouring.

Finally, our scaling relation analysis is based on logarithmic transformations and low dimensional linear models, and therefore primarily probes power law dependencies. More complex functional forms and higher order interactions are not explored given the limited sample size and the associated risk of overfitting. Although most of the relations considered involve only one or two predictors, the number of candidate models tested means that some correlations may still be affected by statistical fluctuations. The derived relations should therefore be regarded as empirical trends, particularly for those with more limited predictive power.

ACKNOWLEDGEMENTS

TF thanks Antti Rantala for a very interesting and enriching discussion. AG acknowledges support from grant ST/Y002385/1. The simulations were run on the Eureka2 HPC cluster at the University of Surrey. This research made use of the following software: AGAMA (Vasiliev 2019), EMCEE (Foreman-Mackey et al. 2013), GRIFIN (Dehnen 2014), JUPYTER (Granger & Perez 2021), MATPLOTLIB (Hunter 2007), NUMPY (Harris et al. 2020), PANDAS (Reback et al. 2020), PYNBODY (Pontzen et al. 2013), PYTHON (Van Rossum & Drake Jr 1995), SCIKIT-LEARN (Pedregosa et al. 2011) and SCIPY (Virtanen et al. 2020).

DATA AVAILABILITY

The authors will share the data underlying this article upon reasonable request.

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APPENDIX A: UNCERTAINTIES ON CORE-SÉRSIC PARAMETERS

We assess the two independent sources of uncertainty affecting our measurements of the break radius R_b and inner logarithmic slope γ from the two-dimensional core-Sérsic fits. For each merger and each parameter, we distinguish the statistical uncertainty arising from the MCMC sampling and the systematic uncertainty arising from the choice of projection.

The MCMC uncertainty is estimated for each projection as the symmetric half-width of the 68% credible interval of the marginalised posterior, $\sigma_{\text{MCMC}} = (\sigma_+ + \sigma_-)/2$, where σ_+ and σ_- are the 84th- and 16th-percentile deviations from the median, respectively. The relative MCMC uncertainty is then $\sigma_{\text{MCMC}}/\hat{\theta}$, where $\hat{\theta}$ denotes the posterior median. We report the median of this quantity over all $N_{\text{proj}} = 12$ projections as the representative MCMC uncertainty for each merger. The projection uncertainty quantifies the sensitivity of the fitted parameter to the viewing angle and is estimated as the standard deviation of the N_{proj} median values across projections, normalised by their median, $\sigma_{\text{proj}} = \text{std}(\hat{\theta}_k)/\text{median}(\hat{\theta}_k)$, where $k = 1, \dots, N_{\text{proj}}$.

Figure A1 summarises the resulting uncertainty distributions across our sample of 28 merger remnants. For the break radius R_b (top row), the MCMC fits are remarkably well constrained, with a median relative uncertainty of only 1.31%. The projection uncertainty is larger, with a median of 4.02%, indicating that the apparent break radius depends non negligibly on the line of sight, a direct consequence of the triaxial morphology of the merger remnants. The median ratio $\sigma_{\text{proj}}/\sigma_{\text{MCMC}} \simeq 4$ confirms that for R_b , the dominant source of uncertainty is the projection rather than the fitting procedure itself.

For the inner slope γ (bottom row), the situation is qualitatively different. The MCMC uncertainty is significantly larger than for R_b , with a median of 7.83%, reflecting the increased difficulty of constraining γ when the central slope is shallow, owing to its covariance with other core-Sérsic parameters. The large relative uncertainties are also partly a consequence of the small values of γ in our remnants, which are typically close to zero, such that even modest absolute

Table A1. Comparison of the fiducial values of the break radius R_b and inner slope γ , obtained from the MCMC fits and measured when the binary semi-major axis reaches $a = a_h/3$, for each merger remnant in the simulations with and without black holes. Fiducial values are defined as the median over projections of the best-fitting values, and the quoted uncertainties correspond to the total uncertainty $\sigma_{\text{tot}} = \sqrt{\sigma_{\text{MCMC}}^2 + \sigma_{\text{proj}}^2}$.

R_b [kpc]		γ	
with BHs	without BHs	with BHs	without BHs
0.221 ± 0.009	0.166 ± 0.008	0.187 ± 0.020	0.162 ± 0.030
0.214 ± 0.006	0.161 ± 0.006	0.140 ± 0.018	0.130 ± 0.030
0.604 ± 0.014	0.505 ± 0.021	0.163 ± 0.016	0.191 ± 0.019
0.467 ± 0.023	0.335 ± 0.014	0.080 ± 0.020	0.004 ± 0.005
0.388 ± 0.043	0.290 ± 0.037	0.080 ± 0.018	0.012 ± 0.010
0.326 ± 0.012	0.264 ± 0.016	0.121 ± 0.021	0.136 ± 0.031
0.199 ± 0.022	0.126 ± 0.027	0.007 ± 0.031	0.004 ± 0.032
1.732 ± 0.078	1.514 ± 0.060	0.122 ± 0.025	0.131 ± 0.055
0.789 ± 0.055	0.655 ± 0.078	0.175 ± 0.012	0.136 ± 0.033
0.801 ± 0.026	0.637 ± 0.077	0.162 ± 0.019	0.236 ± 0.026
1.239 ± 0.089	1.146 ± 0.089	0.004 ± 0.009	0.001 ± 0.017
0.473 ± 0.029	0.384 ± 0.028	0.171 ± 0.013	0.142 ± 0.045
0.920 ± 0.034	0.603 ± 0.025	0.150 ± 0.011	0.114 ± 0.014
0.295 ± 0.027	0.260 ± 0.022	0.133 ± 0.012	0.112 ± 0.011
0.747 ± 0.030	0.649 ± 0.034	0.272 ± 0.014	0.256 ± 0.009
2.001 ± 0.506	1.779 ± 0.468	0.048 ± 0.032	0.008 ± 0.020
0.847 ± 0.022	0.538 ± 0.019	0.144 ± 0.022	0.098 ± 0.038
1.581 ± 0.207	1.387 ± 0.219	0.065 ± 0.064	0.006 ± 0.056
0.971 ± 0.035	0.921 ± 0.048	0.191 ± 0.012	0.182 ± 0.011
0.891 ± 0.023	0.756 ± 0.022	0.140 ± 0.014	0.084 ± 0.032
0.716 ± 0.028	0.496 ± 0.067	0.130 ± 0.021	0.010 ± 0.012
0.959 ± 0.030	0.711 ± 0.022	0.142 ± 0.018	0.097 ± 0.028
0.979 ± 0.022	0.854 ± 0.021	0.115 ± 0.019	0.078 ± 0.030
0.452 ± 0.009	0.349 ± 0.009	0.140 ± 0.015	0.127 ± 0.022
0.939 ± 0.057	0.863 ± 0.048	0.221 ± 0.011	0.195 ± 0.022
0.350 ± 0.065	0.322 ± 0.067	0.008 ± 0.024	0.005 ± 0.013
1.517 ± 0.338	1.626 ± 0.386	0.025 ± 0.071	0.154 ± 0.106
0.711 ± 0.082	0.621 ± 0.104	0.121 ± 0.012	0.138 ± 0.011

uncertainties translate into comparatively large fractional uncertainties. The projection uncertainty is comparable in magnitude, with a median of 10.61%, and the median ratio $\sigma_{\text{proj}}/\sigma_{\text{MCMC}} \simeq 1.43$ shows that the two sources of uncertainty contribute at a similar level for γ . This contrasts with R_b , for which the projection effect clearly dominates. The fiducial values of R_b and γ for each merger remnant, for simulations with and without black holes, are listed in Table A1.

APPENDIX B: TEMPORAL EVOLUTION OF THE CORE-SÉRSIC PARAMETERS

The evolution of the core-Sérsic parameters for five representative mergers evolved beyond $a_h/3$ is shown as a function of $\ln(a_h/a)$ in Fig. B1, where decreasing binary separation corresponds to the progression of time. Each colour represents one projection of a different merger and the error bars indicating the statistical uncertainties from the MCMC fits. Until $a = a_h/5$, the break radius R_b increases systematically as the binary hardens, in agreement with the results of Dosopoulou et al. (2021), who find that R_b grows until approximately this separation and subsequently remains largely unchanged. In contrast, the inner slope γ remains broadly constant throughout the evolution, consistent with the remnants already exhibiting relatively shallow central profiles at t_b . This suggests that stopping our simulations at $a_h/3$ should have little impact on the inferred value

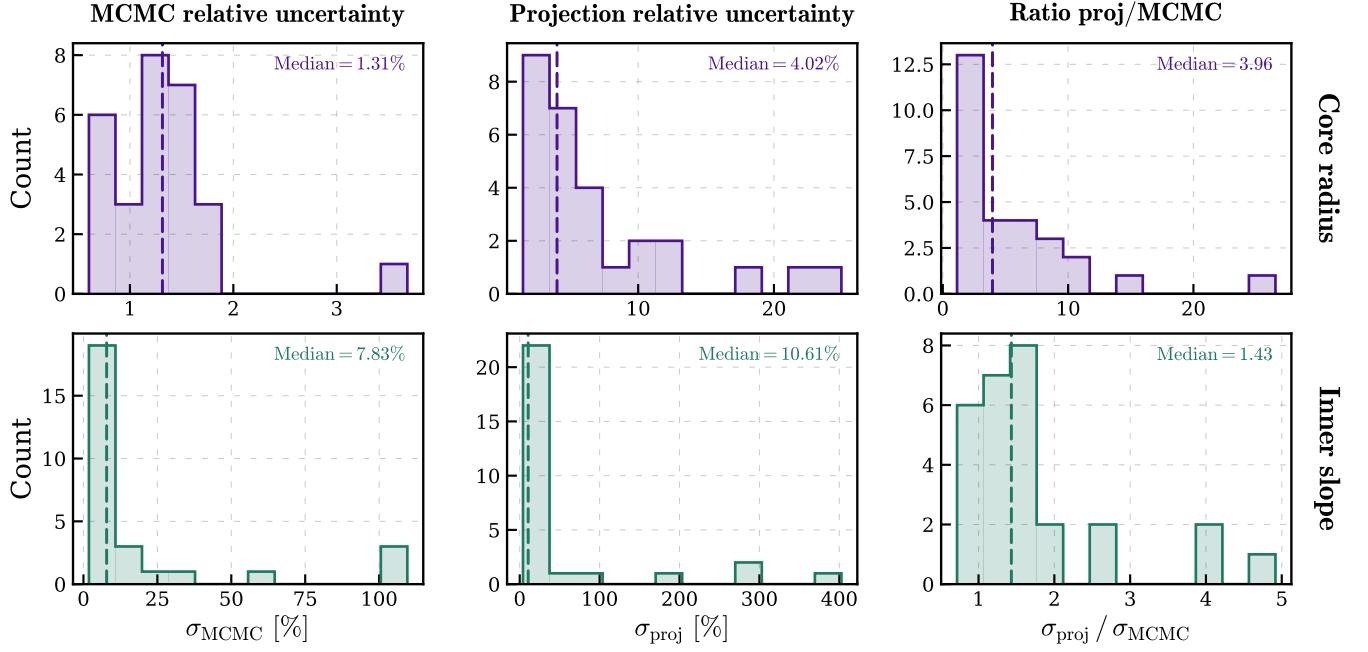


Figure A1. Distributions of the relative uncertainties associated with the two-dimensional core-Sérsic fits of the break radius, R_b (top row), and inner logarithmic slope, γ (bottom row), for the 28 merger remnants. The left column shows the relative uncertainty arising from the MCMC sampling, σ_{MCMC} , the middle column shows the uncertainty associated with the choice of projection, σ_{proj} , and the right column shows their ratio, $\sigma_{\text{proj}}/\sigma_{\text{MCMC}}$. Vertical dashed lines indicate the median values of each distribution, which are reported in each panel. For R_b , the projection uncertainty is systematically larger than the MCMC uncertainty, with median values of 4.02% and 1.31%, respectively. For γ , the two sources of uncertainty are more comparable, with median values of 10.61% and 7.83%, respectively.

of γ , although it may lead to a systematic underestimate of R_b . This latter effect may be less pronounced for highly eccentric binaries, for which gravitational wave emission can become dominant earlier and accelerate the final inspiral, leaving less time for further structural evolution.

APPENDIX C: AXIS RATIOS

Although the intrinsic shape of the remnants is characterised throughout the main text by the triaxiality parameter T , Fig. C1 shows the corresponding axis ratios b/a and c/a at three characteristic scales. For the global shape, measured using the 50% most bound stellar particles (triangles), the two samples largely overlap. This confirms the result discussed in the main text that the presence of SMBHs has little effect on the large-scale intrinsic shape of the merger remnants. The points lie predominantly above the $T = 0.5$ boundary, indicating that most remnants are prolate rather than oblate or maximally triaxial. Only a small number of remnants approach the $T = 0.5$ boundary.

The axis ratios measured on smaller scales show a different behaviour. Within $r_{5\%}$ (diamonds), the two populations begin to separate, with the remnants hosting SMBHs tending to occupy a different region of the $(b/a, c/a)$ plane than those without SMBHs. The separation becomes more pronounced within the binary influence radius, r_{inf} (circles). The central axis ratios are, however, generally close to unity, indicating that the innermost stellar distributions are close to spherical. In this regime, small differences in b/a and c/a can therefore produce relatively large changes in the triaxiality parameter,

making the corresponding central values of T more sensitive to small variations in the axis ratios.

APPENDIX D: THE IMPACT OF GAS CONTENT ON VELOCITY ANISOTROPY

To further assess the impact of gas content on the velocity anisotropy of ILLUSTRISTNG galaxies, we repeat the analysis using progressively more restrictive upper limits on the gas-to-stellar mass ratio. Figure D1 shows the corresponding probability density distributions of β_{hm} for $M_{\text{gas}}/M_{\star} < 0.1, 0.2, 0.3, 0.4$, and 0.5 . The distributions exhibit a systematic shift towards higher β_{hm} values as the maximum allowed gas fraction is decreased, indicating increasingly radial stellar velocity distributions in more gas poor systems. This confirms that gas content contributes to the lower velocity anisotropy measured in ILLUSTRISTNG galaxies, although the relatively modest change in the distributions over most of the explored gas fraction range suggests that it cannot by itself account for the full offset with our gas free merger remnants.

APPENDIX E: INNER SLOPES OF THE PROGENITOR GALAXIES

Figure E1 shows the distributions of the inner density slopes of the first and second progenitor galaxies, both in three dimensions, $\gamma_{\text{pr},3\text{D}}$, and in projection, $\gamma_{\text{pr},2\text{D}}$. The three-dimensional slopes span the range $0.4 \leq \gamma_{\text{pr},3\text{D}} \leq 1.2$, with the first progenitors preferentially clustered around $\gamma_{\text{pr},3\text{D}} \approx 0.5$, while the second progenitors

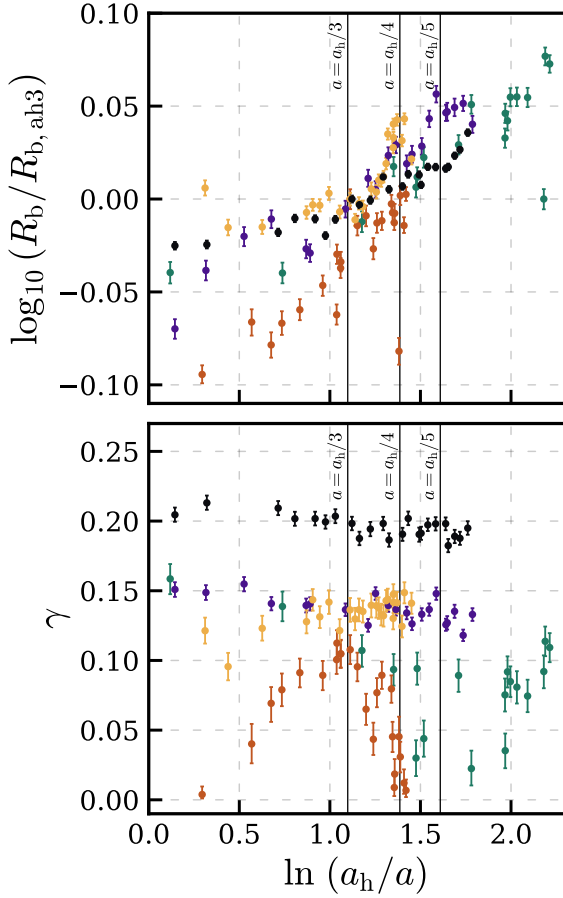


Figure B1. Evolution of the core-Sérsic parameters as a function of the binary separation, expressed as $\ln(a_h/a)$. The top panel shows the break radius R_b , normalised to its value at $a = a_h/3$, while the bottom panel shows the inner logarithmic slope γ . Each colour corresponds to one projection of a different merger, with error bars indicating the statistical uncertainties from the MCMC fits. The vertical lines mark $a = a_h/3$, $a = a_h/4$, and $a = a_h/5$. The break radius increases continuously during the hardening phase, whereas γ remains approximately constant.

exhibit a broader distribution extending towards steeper slopes. Nevertheless, a substantial fraction of both progenitor populations have $\gamma_{\text{pr},3\text{D}} < 1$, and the majority of the systems lie at relatively shallow slopes, $\gamma_{\text{pr},3\text{D}} \lesssim 0.7$. The projected slopes are correspondingly much shallower, spanning approximately $0.1 \lesssim \gamma_{\text{pr},2\text{D}} \lesssim 0.35$, with the first progenitors concentrated near $\gamma_{\text{pr},2\text{D}} \approx 0.1$ – 0.15 and the second progenitors showing a broader distribution extending to $\gamma_{\text{pr},2\text{D}} \approx 0.3$. The progenitors extracted from ILLUSTRISTNG do not generally resemble the steep cusps often adopted in idealised merger simulations.

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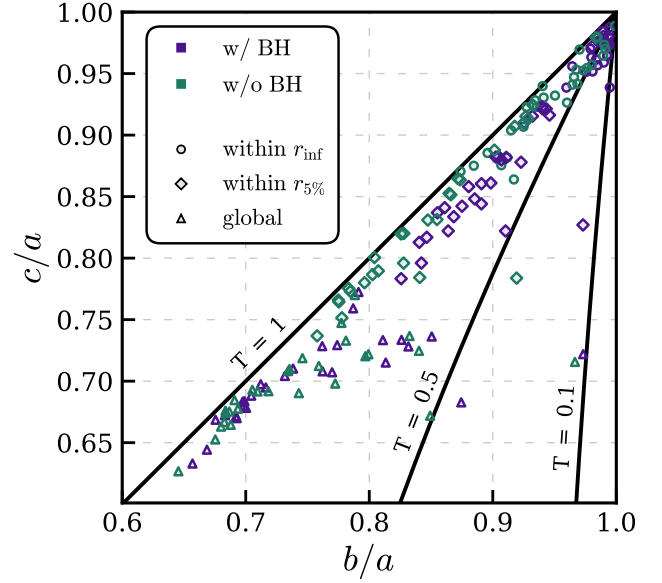


Figure C1. Intrinsic axis ratios of the merger remnants in simulations with SMBHs (purple) and without SMBHs (green), shown in the $(b/a, c/a)$ plane. The black curves indicate lines of constant triaxiality, $T = 1$, 0.5 , and 0.1 , from left to right. Circles, diamonds, and triangles show the axis ratios measured within the binary influence radius r_{inf} , within the radius $r_{5\%}$ enclosing 5% of the total stellar mass, and globally using the 50% most bound stellar particles, respectively. At large scales, the two samples largely overlap, indicating that the presence of SMBHs has little effect on the intrinsic shape of the remnants. The two populations become increasingly separated towards smaller radii, with the strongest difference within r_{inf} . At these central scales, however, the axis ratios are close to unity, making the inferred triaxiality particularly sensitive to small variations in b/a and c/a .

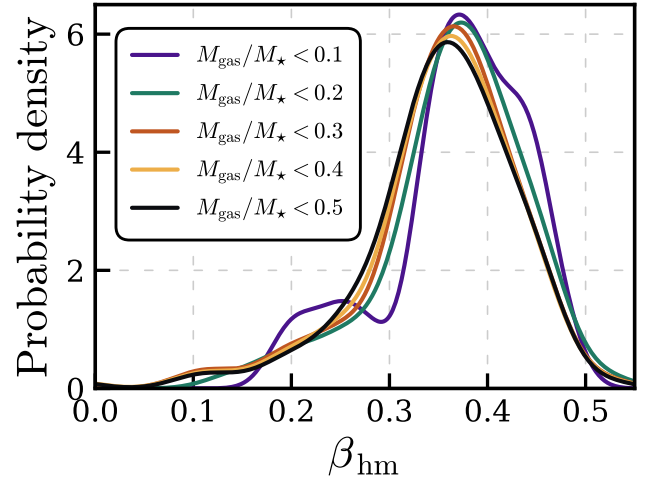


Figure D1. Probability density distributions of the stellar velocity anisotropy parameter, β , measured within the stellar half-mass radius for ILLUSTRISTNG galaxies selected using different upper limits on the gas-to-stellar mass ratio, $M_{\text{gas}}/M_{\star}$. Progressively restricting the sample to lower gas fractions shifts the distribution towards higher β values, indicating increasingly radially anisotropic stellar orbits.

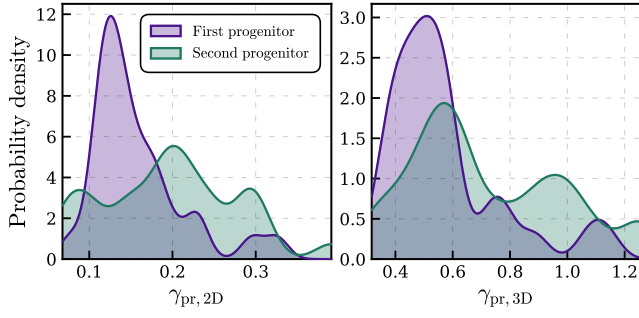


Figure E1. Probability density distributions of the inner density slopes of the progenitor galaxies. The purple and green curves show the distributions for the first and second progenitors, respectively. The left panel shows the inner slopes measured from the projected surface-density profiles, $\gamma_{\text{pr}, 2\text{D}}$, while the right panel shows those measured from the three-dimensional density profiles, $\gamma_{\text{pr}, 3\text{D}}$. The progenitors span a broad range of inner slopes, with most systems exhibiting relatively shallow central profiles.